

Optimal Transfer Rules and Heterogeneity

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Abstract

How should transfers respond to household financial constraints during recessions? This paper studies simple, automatic transfer rules in an estimated two-agent New Keynesian model in which households move endogenously between saver and hand-to-mouth states. The analysis yields two main findings. First, allowing household composition to respond to economic conditions improves the model’s empirical fit relative to a standard two-agent model. Second, a rule responding to output and the hand-to-mouth share of aggregate consumption—a measure reflecting both relative consumption and household composition—most closely approximates the Ramsey allocation under both tax and debt financing. Applied to the U.S. COVID-19 recession, the rule would have front-loaded smaller transfers with a gradual phase-out as the recovery unfolded, stabilizing household income at considerably lower fiscal cost.

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1 Introduction

Throughout the early 2000s recession, the global financial crisis, and the COVID-19 pandemic, the U.S. Congress increasingly prioritized direct financial assistance to low-income households as part of its economic response. These payments, distributed as stimulus checks of up to \$1,400 per eligible individual, represented between 3% and 19% of the government’s current transfer payments, totaling a record \$814 billion in 2020 (see the left panel in Figure 1). While designed to alleviate economic hardship and stimulate spending during downturns, these discretionary measures face significant challenges. Fiscal policy lags, including delays in decision-making and implementation, often undermine their timeliness. Moreover, determining the appropriate size of transfers poses an additional challenge. Overly large payments may induce a substantial saving response, while insufficient payments may fail to stabilize demand or support vulnerable households.

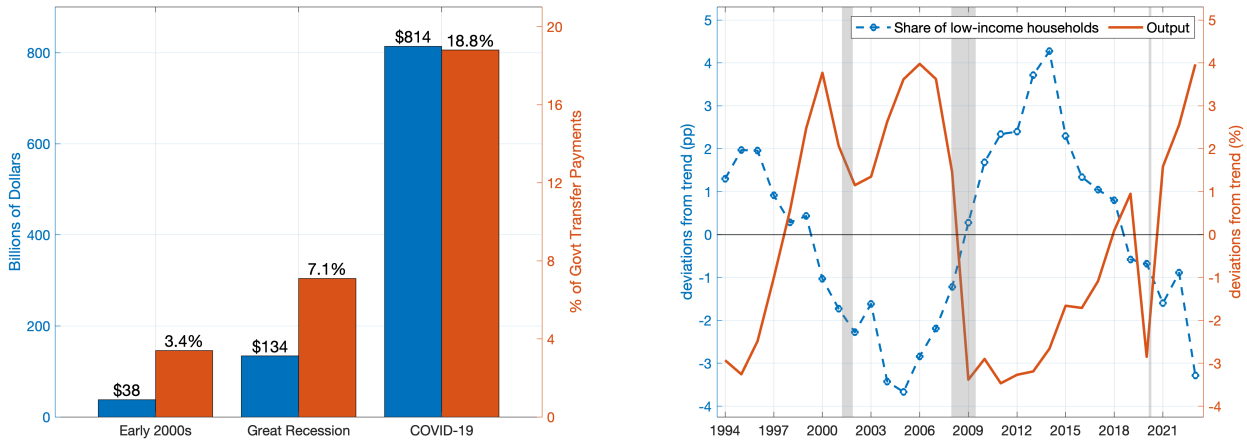


Figure 1: Stimulus checks and household income during economic downturns

(Left panel) Tax rebates and economic payments were provided in response to major economic downturns: the 2001 recession (Economic Growth and Tax Relief Reconciliation Act), the Great Recession (Economic Stimulus Act of 2008 and American Recovery and Reinvestment Act of 2009), and the COVID-19 pandemic (CARES Act, Tax Relief Act of 2020, and American Rescue Plan Act of 2021). Stimulus checks are shown as a share of government transfer payments during these periods. Source: *congress.gov*. (Right panel) Quadratically detrended share of households with before-tax family income below 200% of the federal poverty guidelines, classified as low-income, as reported in the Consumer Expenditure Survey. The output gap is defined as the log difference between observed and potential real GDP (CBO estimates). See Appendix D.1 for more details.

Against this backdrop, this paper examines how simple, automatic rules can improve the timing, size, and targeting of fiscal transfers to provide a more effective response to prevailing

economic conditions. The analysis employs an otherwise-standard two-agent New Keynesian model extended to allow for endogenous switching between household types. This extension allows a joint analysis of the role of fiscal transfers in mitigating cyclical income risk among financially constrained households—a margin absent from conventional two-agent models—and in stimulating consumption. Specifically, the model introduces time-varying probabilities into the tractable HANK (THANK) model of Bilbiie (2024), governing transitions into and out of the financially constrained state. These probabilities respond endogenously to changes in disposable income, in turn generating time variation in the relative shares of constrained and unconstrained households.

The proposed *dynamic* THANK (D-THANK) model retains the tractability of the two-agent framework while incorporating elements from models featuring richer heterogeneity (e.g., HANK). It introduces a novel composition channel through which aggregate dynamics are shaped not only by the distinct consumption and saving behavior of constrained and unconstrained households but also by changes in the relative proportions of the two types. The endogenous variation in household composition is consistent with the extensive empirical evidence on the countercyclical nature of income risk (Storesletten et al., 2004; Guvenen et al., 2014) and is further motivated by the evidence presented in the right panel of Figure 1. The figure shows that the share of households classified as low-income varies systematically with economic conditions,¹ increasing during downturns and decreasing during expansions.

The D-THANK model also incorporates into the Ramsey planner’s welfare objective a distributional margin emphasized in richer heterogeneous-agent models (e.g., McKay and Wolf, 2023), in which consumption dispersion reflects both differences in consumption across household types and endogenous changes in household composition. The planner therefore accounts for welfare losses arising from cyclical income risk among financially constrained

¹I classify households with family income below 200% of the federal poverty guidelines as low-income. This threshold matches the model’s steady-state share of constrained households and is consistent with the income levels of hand-to-mouth households documented by Aguiar et al. (2024). Appendix D.1 details the classification and shows that the resulting series displays very similar cyclical dynamics across alternative thresholds.

households—a dimension absent from models that assume a fixed household distribution.

Two key findings emerge from the analysis. First, by incorporating endogenous changes in household composition, the D-THANK model provides a better empirical fit than the THANK model, as evidenced by several model selection criteria. Second, across both tax- and debt-financing regimes, a simple transfer rule responding to output and the consumption share of hand-to-mouth households minimizes the welfare distance from the Ramsey allocation among the rules considered. The policy relevance of the consumption share stems from its ability to summarize distributional conditions through a single indicator that reflects both differences in relative consumption and endogenous changes in household composition.

The quantitative implications of these findings are further illustrated through a counterfactual analysis of the COVID-19 recession, assuming the U.S. government had implemented the proposed consumption-share transfer rule. The counterfactual underscores the importance of conditioning fiscal support on prevailing economic conditions. At the onset of the pandemic, the rule calls for a rapid fiscal response but prescribes a substantially smaller intervention—approximately 3% of GDP, compared with roughly 10% under the first round of actual payments. As economic conditions improve, the rule calls for a gradual withdrawal of support, in contrast to the second large round of transfers during the recovery, which raised household income to nearly 12% above its pre-pandemic level.²

Related literature. This paper contributes to the growing literature that integrates micro-heterogeneity channels from quantitative HANK models into analytically tractable frameworks (e.g., Bilbiie, 2024; Debortoli and Gali, 2024; Ravn and Sterk, 2021). Building on the two-agent framework of Bilbiie (2024), the D-THANK model incorporates a novel mechanism that allows households to switch types endogenously, with the relative shares of each type adjusting dynamically in response to aggregate shocks. A closely related contribution is Bilbiie et al. (2023), who model time-varying switching probabilities only for un-

²Using a large-scale consumer survey, Coibion et al. (2020) find that households spent approximately 40 percent of their stimulus payments, while saving the remainder or using it to repay debt, suggesting that the payments did not translate proportionately into additional consumption.

constrained households, emphasizing cyclical precautionary saving as a source of economic amplification while keeping the aggregate shares of household types fixed. The D-THANK framework instead allows *both* constrained and unconstrained households to switch types endogenously. In this way, the model captures endogenous shifts in household composition that are characteristic of quantitative HANK models (e.g., Kaplan et al., 2018; McKay and Wolf, 2023; Bayer et al., 2023; LeGrand et al., 2024), while avoiding the computational complexity associated with tracking time-varying income distributions in the presence of idiosyncratic shocks and high-dimensional state spaces.

This paper also contributes to the literature on optimal fiscal policy in heterogeneous-agent models. In this regard, Bilbiie et al. (2024) study monetary and fiscal policy in a standard two-agent New Keynesian (TANK) framework, while Wu and Xie (2024) extend this analysis to unconventional monetary policy. In these frameworks, the welfare costs of cross-sectional consumption dispersion stem entirely from consumption differences across household types (i.e., the consumption gap), given a fixed distribution of households. In contrast, the D-THANK model relaxes this assumption and broadens the Ramsey planner’s problem to account for changes in the shares of households across income groups. This additional margin introduces cyclical income risk as a distinct source of welfare volatility, rendering strict consumption-gap targeting suboptimal.³

A third contribution concerns the design of automatic fiscal rules. McKay and Reis (2016) quantify the extent to which the U.S. tax-and-transfer system stabilizes aggregate fluctuations, while McKay and Reis (2021) study the optimal design of these automatic stabilizers, focusing specifically on tax progressivity and transfer benefits. In both analyses, however, the strength of fiscal stabilization is fixed and does not vary systematically with the state of the business cycle (i.e., aggregate support rises in downturns because more households qualify for benefits, while the benefit schedule itself remains unchanged). This

³A related literature studies the distributional effects of monetary policy and proposes augmented Taylor rules that explicitly respond to distributional considerations (e.g., Yang, 2023). In contrast, this paper derives fiscal rules that are *conditional* on the central bank’s mandate, as defined by a standard Taylor rule.

paper instead considers a complementary policy dimension in which fiscal transfers respond dynamically to macroeconomic and distributional conditions (i.e., the size of the transfer received by each constrained household adjusts to the severity of the downturn). A further distinction concerns eligibility, which in this framework is based on financial constraints rather than unemployment status, consistent with the design of U.S. stimulus payments directed at income-eligible households regardless of employment status. In this respect, the analysis is closely related to McKay and Wolf (2023), who characterize optimal fiscal transfers in a quantitative HANK model. The D-THANK framework complements their approach by deriving the optimal path of fiscal transfers under the Ramsey allocation and using it as a normative benchmark for the design and evaluation of simple, implementable transfer rules based on observable economic indicators. More broadly, related work on systematic fiscal rules includes Kliem and Kriwoluzky (2014), who study rules for capital and labor income taxation, and Evers (2012), who analyzes rules governing federal fiscal transfers in monetary unions.

Finally, this paper contributes to the literature on fiscal stimulus payments during the COVID-19 recession. While existing studies, including Bayer et al. (2023) and Faria-e-Castro (2021), quantify the effects of fiscal stimulus during the pandemic, this paper uses the COVID-19 episode as a laboratory to compare economic losses across counterfactual policy scenarios and quantify the potential gains from a simple state-contingent transfer rule.

The remainder of the paper is organized as follows. Section 2 introduces the D-THANK model. Section 3 describes the empirical strategy and presents the estimation results. Section 4 derives the welfare criterion, characterizes the Ramsey allocation, and evaluates simple transfer rules under alternative financing regimes. Section 5 presents the COVID-19 counterfactual analysis. The final section concludes.

2 The dynamic THANK model

The dynamic THANK (D-THANK) model builds on the tractable two-agent framework of Bilbiie (2024), incorporating endogenous switching between household types and allowing their proportions to evolve dynamically over time.

The economy consists of two distinct types of agents: hand-to-mouth (H) agents and savers (S), whose population shares are denoted λ_t and $1 - \lambda_t$, respectively. Hand-to-mouth agents are financially constrained and must consume their entire disposable income each period, while savers are able to smooth consumption over time by investing in risk-free bonds. In each period, a fraction of households transitions between types according to a *time-varying* Markov process, which adjusts dynamically in response to fluctuations in disposable labor income. Agents anticipate these fluctuations and self-insure based on the probability of such events occurring. As a result, consumption–saving decisions and optimal labor supply choices are made after aggregate shocks occur but before household types are determined.

In line with Schmitt-Grohe and Uribe (2005), households supply homogeneous labor to labor unions, which then transform it into differentiated labor inputs and sell these to intermediate firms. Labor unions and intermediate firms face nominal rigidities in their wage-setting and price-setting decisions, respectively. Additionally, the central bank sets the nominal interest rate according to a standard Taylor rule, while the government provides lump-sum transfers to financially constrained households. In the baseline model, these transfers are exogenous and financed in each period through taxes on savers. Section 4 extends the framework by introducing transfer rules that respond endogenously to macroeconomic conditions and by allowing government debt issuance as an alternative to balanced-budget financing.

2.1 Households

Each household type $i \in \{H, S\}$ derives utility from consumption C_t^i and disutility from labor supply N_t^i , and can save through risk-free bonds Z_{t+1}^i . Preferences are represented by the discounted sum of utility:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(\log(C_t^i) - \frac{(N_t^i)^{1+\varphi}}{1+\varphi} \right) \quad (2.1)$$

where $\beta \in (0, 1)$ is the subjective discount factor, $C_t^i \equiv \left(\int_0^1 C_t^i(j)^{\frac{\varepsilon_p-1}{\varepsilon_p}} dj \right)^{\frac{\varepsilon_p}{\varepsilon_p-1}}$ is an index of intermediate consumption goods $C_t^i(j)$, $\varepsilon_p > 1$ represents the price elasticity of substitution, and φ is the inverse Frisch elasticity.

At the individual level, agents transition between state H and state S according to an endogenous transition matrix:

$$P = \begin{bmatrix} s_t & 1 - s_t \\ 1 - h_t & h_t \end{bmatrix} \quad (2.2)$$

where s_t and h_t represent the probabilities that savers and hand-to-mouth agents, respectively, remain in their current states. A more detailed explanation of the process governing the dynamics of s_t and h_t is provided below.

For each individual of type i , the period budget constraints are:

$$(H) : \quad C_t^H + Z_{t+1}^H = \frac{W_t}{P_t} N_t^H + T_t^H + \frac{1 + i_{t-1}}{1 + \pi_t} \frac{\mathcal{B}_t^H}{\lambda_t} \quad (2.3)$$

$$(S) : \quad C_t^S + Z_{t+1}^S = \frac{W_t}{P_t} N_t^S + T_t^S + \frac{1}{1 - \lambda_t} D_t + \frac{1 + i_{t-1}}{1 + \pi_t} \frac{\mathcal{B}_t^S}{1 - \lambda_t} \quad (2.4)$$

These constraints are defined in terms of real labor income $\frac{W_t}{P_t} N_t^i$, lump-sum taxes/transfers T_t^i , and dividends D_t from firms owned by savers, where $\frac{W_t}{P_t}$ is the real wage. As equity

is held exclusively by savers, the entire dividend stream accrues to them and is absent from the hand-to-mouth budget constraint, leaving lump-sum transfers as the only redistributive instrument that reaches constrained households.⁴

Additionally, they include accumulated financial wealth $\mathcal{B}_{t+1}^i$, which results from the aggregation of savings Z_{t+1}^i by agents transitioning to the same state:

$$\mathcal{B}_{t+1}^H = \lambda_t h_{t+1} Z_{t+1}^H + (1 - \lambda_t) (1 - s_{t+1}) Z_{t+1}^S \quad (2.5)$$

$$\mathcal{B}_{t+1}^S = \lambda_t (1 - h_{t+1}) Z_{t+1}^H + (1 - \lambda_t) s_{t+1} Z_{t+1}^S \quad (2.6)$$

These are remunerated at the gross ex post real interest rate $\frac{1+i_{t-1}}{1+\pi_t}$, where i_{t-1} is the nominal interest rate from the previous period and π_t the rate of inflation. To maintain tractability, the original model by Bilbiie (2024) imposes a series of additional assumptions regarding agents' ability to save and borrow, which are discussed in further detail below.

Optimality conditions

Let there be a family head who optimally allocates resources on behalf of savers and hand-to-mouth agents according to the value function:

$$V(\mathcal{B}_t^S, \mathcal{B}_t^H) = \max_{\substack{C_t^S, Z_{t+1}^S, \\ C_t^H, Z_{t+1}^H}} \{ (1 - \lambda_t) \log C_t^S + \lambda_t \log C_t^H + \beta E_t [V(\mathcal{B}_{t+1}^S, \mathcal{B}_{t+1}^H)] \} \quad (2.7)$$

subject to the budget and wealth-accumulation constraints (2.3)-(2.6) and the borrowing constraints $Z_{t+1}^S \geq 0, Z_{t+1}^H \geq 0$.⁵

⁴This specification corresponds to the case $\tau^D = 0$ in the redistribution framework of Bilbiie (2024) and Bilbiie et al. (2021), where τ^D is the fraction of dividend income rebated to hand-to-mouth households. Restricting equity income to savers reflects the empirical concentration of asset ownership at the top of the wealth distribution, consistent with hand-to-mouth households holding little or no equity (Aguar et al., 2024; Bilbiie, 2024; Heathcote et al., 2020).

⁵As in Schmitt-Grohe and Uribe (2005), labor is supplied through utility-maximizing unions that act on behalf of households, with hours determined by firms' labor demand at the wage set by the unions. Consequently, labor supply does not enter the household optimization problem and is instead characterized within the wage-setting block described below.

Optimal consumption/saving decisions are described by the following Euler equations:

$$(C_t^S)^{-1} = \beta E_t \left[\frac{1+i_t}{1+\pi_{t+1}} \left(s_{t+1} (C_{t+1}^S)^{-1} + (1-s_{t+1}) (C_{t+1}^H)^{-1} \right) \right] + \Xi_t^S \quad (2.8)$$

$$(C_t^H)^{-1} = \beta E_t \left[\frac{1+i_t}{1+\pi_{t+1}} \left(h_{t+1} (C_{t+1}^H)^{-1} + (1-h_{t+1}) (C_{t+1}^S)^{-1} \right) \right] + \Xi_t^H \quad (2.9)$$

where $\Xi_t^i \geq 0$ for $i \in \{H, S\}$ are the Kuhn–Tucker multipliers associated with the borrowing constraints $Z_{t+1}^i \geq 0$. To maintain tractability, three simplifying assumptions are introduced in Bilbiie (2024). First, it is assumed that savers will never reach the borrowing limit, i.e., $\Xi_t^S = 0$ and $Z_{t+1}^S \neq 0$ for all t . Second, hand-to-mouth agents are assumed to be financially constrained in every period, i.e., $\Xi_t^H \neq 0$ and $Z_{t+1}^H = 0$ for all t , and they must consume their entire period’s disposable income. Third, a zero-liquidity limit is imposed on savers’ budget constraint, meaning that bonds are in zero net supply in equilibrium. This final assumption is relaxed in Section 4, which examines the implications of debt-financed fiscal transfers. As a result, the optimal consumption decisions for savers and hand-to-mouth agents reduce to:

$$(C_t^S)^{-1} = \beta E_t \left[\frac{1+i_t}{1+\pi_{t+1}} \left(s_{t+1} (C_{t+1}^S)^{-1} + (1-s_{t+1}) (C_{t+1}^H)^{-1} \right) \right] \quad (2.10)$$

$$C_t^H = \frac{W_t}{P_t} N_t^H + T_t^H \equiv Y_t^D \quad (2.11)$$

Here, savers account for the possibility of transitioning to the hand-to-mouth state by “self-insuring” in the present. Conversely, a hand-to-mouth household is constrained to consume its disposable labor income Y_t^D , namely labor earnings plus net transfers.

Time-varying probabilities. The transition probabilities are allowed to vary over time around their steady-state values, s and h , according to an error-correction process that depends on disposable labor income and the composition of households:⁶

⁶Bilbiie et al. (2023) allow only the probability that savers remain in the saver state, s_t , to vary over time. In their specification, s_t is modeled as a function of the GDP gap and does not include an error-correction

$$h_t = h \left(\frac{Y_t^D}{Y^D} \right)^{-\gamma_h} e^{-\kappa_h(\lambda_{t-1}-\lambda)} \quad (2.12)$$

$$s_t = s \left(\frac{Y_t^D}{Y^D} \right)^{\gamma_s} e^{\kappa_s(\lambda_{t-1}-\lambda)} \quad (2.13)$$

where $\gamma_h, \gamma_s \geq 0$ govern the sensitivity of transition probabilities to fluctuations in disposable labor income. That is, an increase in disposable labor income relative to its steady-state level raises the probability that savers remain in the saver state and lowers the probability that hand-to-mouth households remain constrained.

This specification is motivated by two considerations. First, disposable labor income constitutes a substantial share of total disposable income for low- and middle-income households, as emphasized by Aguiar et al. (2024). Consequently, within the model, changes in labor earnings are treated as uninsurable income shocks, leading to short-term fluctuations in the share of hand-to-mouth individuals.⁷ This mechanism is also consistent with the empirical analysis of Heathcote et al. (2020), who find that the countercyclicality of income inequality is primarily driven by dynamics at the lower end of the earnings distribution. Second, fiscal stimulus payments create a wealth effect through households' budget constraints, effectively functioning as an income shock. This channel aligns with empirical observations from the COVID-19 recession, where fiscal transfers helped limit the rise in the poverty rate by approximately 3.6% (Burns et al., 2021). Additionally, quantitative analyses by Bayer et al. (2023) indicate that fiscal transfers effectively contained the increase in the income Gini coefficient by approximately 3%.

While the income channel allows household composition to respond endogenously to changes in economic conditions, temporary income fluctuations should not generate permanent shifts in the distribution of households across liquidity states. To capture this feature, the parameters $\kappa_s, \kappa_h \geq 0$ govern the speed at which the household distribution returns

component, corresponding to the special case $\kappa_s = \kappa_h = 0$.

⁷In models featuring rich heterogeneity, these fluctuations resemble the direct income effects on households near the borrowing constraint.

to its steady state. Specifically, the transition probabilities depend on deviations of the constrained-household share, λ_{t-1} , from its long-run value, λ . This error-correction mechanism ensures that the distribution remains anchored to its ergodic benchmark while providing additional flexibility in the estimation of the model.

At the aggregate level, the shares of savers $(1 - \lambda_t)$ and hand-to-mouth agents (λ_t) also evolve over time, following a Markov process:

$$\begin{bmatrix} 1 - \lambda_t & \lambda_t \end{bmatrix} = \begin{bmatrix} 1 - \lambda_{t-1} & \lambda_{t-1} \end{bmatrix} \begin{bmatrix} s_t & 1 - s_t \\ 1 - h_t & h_t \end{bmatrix} \quad (2.14)$$

In log-linear form, the process is expressed as:

$$\hat{\lambda}_t = (s + h - 1) \hat{\lambda}_{t-1} + \lambda \hat{h}_t - (1 - \lambda) \hat{s}_t. \quad (2.15)$$

When combined with the log-linearized Equations (2.12) and (2.13), this yields:

$$\hat{\lambda}_t = \rho_\lambda \hat{\lambda}_{t-1} - (\tilde{\gamma}_s + \tilde{\gamma}_h) \hat{y}_t^D \quad (2.16)$$

Here, $\tilde{\gamma}_s \equiv (1 - \lambda) \cdot s \cdot \gamma_s$, $\tilde{\gamma}_h \equiv \lambda \cdot h \cdot \gamma_h$, and $\rho_\lambda \equiv (s + h - 1) - (1 - \lambda) \cdot s \cdot \kappa_s - \lambda \cdot h \cdot \kappa_h$. Therefore, conditional on $\tilde{\gamma}_s > 0$ and $\tilde{\gamma}_h > 0$, an increase in disposable labor income results in a decrease in the aggregate share of hand-to-mouth agents and an increase in the share of savers in the economy.

Importantly, the D-THANK model expands on the household blocks of the THANK models developed in Bilbiie (2024) and Bilbiie et al. (2023). When both $\tilde{\gamma}_s$ and $\tilde{\gamma}_h$ are set to zero, the model simplifies to that of Bilbiie (2024), where the Markov process in (2.14) is in equilibrium, and there is no time variation in the probabilities of transitioning between household types or in their distribution. Similarly, by fixing the transition probability h_t and the distribution of household types at their steady-state values, and setting $\kappa_s = \kappa_h = 0$, the household block of the model reduces to that of Bilbiie et al. (2023).

Labor supply. As in Schmitt-Grohe and Uribe (2005), households supply different types of labor, indexed by $z \in [0, 1]$. Type-specific unions aggregate labor inputs and exchange them with firms for a nominal wage $W_t(z)$. Given $W_t(z)$, a worker of type z supplies labor $N_t(z)$ as demanded by firms:

$$N_t(z) = \left(\frac{W_t(z)}{W_t} \right)^{-\varepsilon_w} N_t^D \quad (2.17)$$

where $\varepsilon_w > 1$ represents the labor elasticity of substitution, W_t is the nominal wage index, and N_t^D denotes the aggregate labor demand.

Households are uniformly distributed across unions, leading to a uniform aggregate demand for labor of type z . As a result, the individual labor supply, $N_t(z)$, is equal across households, such that $N_t^S = N_t^H = N_t$.

2.2 Labor markets

Nominal wages are sticky and can be adjusted with a probability of $1 - \theta_w$ in each period. As described in Colciago (2011) and Gali et al. (2007), wages are set by labor unions, which seek to maximize a social welfare function that averages the utility of hand-to-mouth households and savers, subject to the labor demand for type z by intermediate firms:

$$\begin{aligned} \max_{W_t^*(z)} E_t \sum_{k=0}^{\infty} (\beta \theta_w)^k & \left[\left(\frac{\lambda_{t+k}}{C_{t+k}^H} + \frac{1 - \lambda_{t+k}}{C_{t+k}^S} \right) \frac{W_t^*(z)}{P_{t+k}} N_{t+k}(z) - \frac{N_{t+k}^{1+\varphi}(z)}{1+\varphi} \right] \\ & s.t. \\ N_{t+k}(z) &= \left(\frac{W_t^*(z)}{W_{t+k}} \right)^{-\varepsilon_w} N_{t+k}^D \end{aligned} \quad (2.18)$$

Importantly, the weights in the social welfare function correspond to the *time-varying* shares of households, λ_{t+k} and $1 - \lambda_{t+k}$, linking changes in the household distribution to wage inflation. In the absence of wage rigidities, the optimal wage simplifies to the condition

derived in Gali et al. (2007):

$$\frac{W_t^*(z)}{P_t} = \mathcal{M}_w \left[\frac{\lambda_t}{MRS_t^H(z)} + \frac{1 - \lambda_t}{MRS_t^S(z)} \right]^{-1} \quad (2.19)$$

where $MRS_t^i(z) \equiv (N_t^i(z))^\varphi C_t^i$ denotes the marginal rate of substitution for agent i and $\mathcal{M}_w \equiv \frac{\varepsilon_w}{\varepsilon_w - 1} > 1$ represents the wage markup. Under the condition that $\frac{W_t^*(z)}{P_t} > MRS_t^S(z) > MRS_t^H(z)$, both households always supply the amount of labor, $N_t(z)$, demanded by the labor unions. As a result, holding everything else constant, an increase in λ_t , reflecting a higher share of constrained households, alters the average marginal rate of substitution, leading to a decline in the nominal wage.

2.3 Production

There exists a continuum of monopolistically competitive firms that produce differentiated intermediate goods. These goods are then utilized as inputs by a perfectly competitive firm to manufacture a single final good.

Final good producer. The representative final good producer uses a constant elasticity of substitution (CES) technology to produce the final good, Y_t :

$$Y_t = \left(\int_0^1 Y_t(j)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} dj \right)^{\frac{\varepsilon_p}{\varepsilon_p - 1}} \quad (2.20)$$

where $Y_t(j)$ denotes the quantity of intermediate good j used as an input.

Profit maximization yields the demand schedule for intermediate good j :

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\varepsilon_p} Y_t \quad (2.21)$$

where $P_t(j)$ and P_t denote the prices of intermediate and final goods, respectively, and are taken as given.

Intermediate goods producers. Labor is the only input used in the production of intermediate goods. The production function is given by:

$$Y_t(j) = A_t N_t(j)^{1-\alpha} \quad (2.22)$$

where α represents the share of capital;⁸ A_t , which represents aggregate productivity, evolves exogenously according to:

$$\log A_t = \rho_a \log A_{t-1} + u_t^a \quad (2.23)$$

where u_t^a is an i.i.d. productivity shock.

Nominal prices are sticky and can be adjusted with probability $1 - \theta_p$ in each period. Firms that are able to adjust their prices in a given period set $P_t^*(j)$ to maximize their expected profits, subject to the demand schedule for good j :

$$\begin{aligned} \max_{P_t^*(j)} E_t \sum_{k=0}^{\infty} \theta_p^k \Lambda_{t,t+k}^S \{P_t^*(j) Y_{t+k}(j) - \Psi(Y_{t+k}(j))\} \\ s.t. \\ Y_{t+k}(j) = \left(\frac{P_t^*(j)}{P_{t+k}} \right)^{-\varepsilon_p} Y_{t+k} \end{aligned} \quad (2.24)$$

where $\Psi(\cdot)$ represents the total costs associated with producing $Y_{t+k}(j)$ units of the good and $\Lambda_{t,t+k}^S \equiv \beta^k (C_{t+k}^S / C_t^S)^{-1}$ is the saver's stochastic discount factor. The optimal reset price, $P_t^*(j)$, satisfies the following condition:

$$\frac{P_t^*(j)}{P_{t-1}} = \mathcal{M}_p \frac{E_t \sum_{k=0}^{\infty} \theta_p^k \Lambda_{t,t+k}^S Y_{t+k}(j) [MC_{t+k}(j) \Pi_{t-1,t+k}]}{E_t \sum_{k=0}^{\infty} \theta_p^k \Lambda_{t,t+k}^S Y_{t+k}(j)} \quad (2.25)$$

where $\mathcal{M}_p \equiv \frac{\varepsilon_p}{\varepsilon_p - 1} > 1$ is the desired price markup, $\Pi_{t,t+k} \equiv P_{t+k}/P_t$ is gross inflation between t and $t+k$, and $MC_{t+k}(j) \equiv \psi_{t+k}(j)/P_{t+k}$ is the real marginal cost. Moreover,

⁸Alternatively, aggregate productivity can be expressed as $A_t \equiv \tilde{A}_t K(j)^\alpha$, where $K(j)$ denotes a fixed capital input in the production function.

cost minimization implies that all firms share the same real marginal cost:

$$MC_t = \frac{W_t}{P_t (1 - \alpha) A_t N_t^{-\alpha}} \quad (2.26)$$

2.4 Monetary policy

A central bank sets the nominal interest rate in each period following a simple Taylor rule

$$i_t = \rho_i i_{t-1} + (1 - \rho_i) [\phi_\pi \pi_t + \phi_y (\hat{y}_t - \hat{y}_{t-1})] + u_t^v \quad (2.27)$$

where ρ_i is the persistence of the nominal interest rate, $\phi_\pi > 0$ and $\phi_y > 0$ are the policy coefficients for inflation and output growth, respectively, $\hat{y}_t$ denotes the deviation of output from its steady state, and u_t^v is an i.i.d. shock to the interest rate.

2.5 Fiscal policy

The government imposes taxes on savers, $\mathcal{T}_t^S \equiv (1 - \lambda_t) T_t^S$, and pays these out as lump-sum transfers to hand-to-mouth agents, $\mathcal{T}_t^H \equiv \lambda_t T_t^H$. In the baseline model, transfers are in zero net supply, so that fiscal policy is balanced in each period:

$$T_t^H \lambda_t + T_t^S (1 - \lambda_t) = 0 \quad (2.28)$$

To reflect their historically sporadic use, lump-sum transfers are assumed to remain at their (zero) steady-state value in the estimation of the D-THANK model. Optimal transfer rules, together with the alternative of debt financing, are introduced in Section 4, where the balanced-budget condition above is relaxed.

2.6 Aggregation and market clearing

This subsection concludes the description of the D-THANK model by presenting the aggregate resource constraints and market clearing conditions. A complete set of nonlinear and

log-linear equilibrium conditions is provided in Appendices A and C.

Aggregate production. Aggregating output across intermediate producers, total production satisfies:

$$Y_t v_t^p = A_t N_t^{1-\alpha} \quad (2.29)$$

where the degree of price dispersion is

$$v_t^p \equiv \int_0^1 \left(\frac{P_t(j)}{P_t} \right)^{-\varepsilon_p} dj. \quad (2.30)$$

Aggregate consumption. Aggregate consumption is calculated as the weighted average of consumption from both households:

$$C_t^A = \lambda_t C_t^H + (1 - \lambda_t) C_t^S \quad (2.31)$$

Price and wage dynamics. All firms and unions re-optimizing in a given period solve identical problems and therefore choose common reset values, so that $P_t^*(j) = P_t^*$ and $W_t^*(z) = W_t^*$. As a result, under Calvo price and wage setting, the aggregate price and wage indices evolve over time according to:

$$P_t^{1-\varepsilon_p} = \theta_p P_{t-1}^{1-\varepsilon_p} + (1 - \theta_p) (P_t^*)^{1-\varepsilon_p} \quad (2.32)$$

$$W_t^{1-\varepsilon_w} = \theta_w W_{t-1}^{1-\varepsilon_w} + (1 - \theta_w) (W_t^*)^{1-\varepsilon_w} \quad (2.33)$$

Labor market. Aggregating individual labor supply over unions, the labor resource constraint is:

$$N_t = N_t^D v_t^w \quad (2.34)$$

where the degree of wage dispersion is

$$v_t^w \equiv \int_0^1 \left(\frac{W_t(z)}{W_t} \right)^{-\varepsilon_w} dz. \quad (2.35)$$

Goods market. Goods market clearing requires $Y_t(j) = C_t(j)$ for all j , and in the aggregate:

$$Y_t = C_t^A \quad (2.36)$$

3 Model estimation and analysis

This section estimates the D-THANK model and the THANK model of Bilbiie (2024). Given the parsimonious structure of both models, the estimation primarily focuses on identifying the structural parameters governing nominal rigidities, monetary policy, and the persistence and volatility of the exogenous disturbances from a small set of aggregate observables. These estimates provide the empirical foundation for the optimal-policy and counterfactual exercises of Sections 4 and 5. Additionally, this section examines the impact of introducing an endogenous household distribution on the model’s empirical fit.

Data and estimation. Both models are estimated with Bayesian methods on U.S. quarterly data for the log difference of real GDP per capita, log hours worked per capita, the log difference of the GDP deflator, the Federal funds rate, and the log difference of nominal wages.⁹ A full description of the data is provided in Appendix D.1.

⁹Because both models describe a closed economy with a single domestically produced and consumed good, the price of the consumption basket coincides with the deflator of domestic value added, making the GDP deflator the natural empirical counterpart. A consumption-based index such as the CPI, by contrast, also reflects import prices, which have no analogue in the model. This choice follows standard practice in estimated closed-economy models, including Smets and Wouters (2007) and Bilbiie et al. (2023).

The corresponding measurement equation is:

$$X_t = \begin{bmatrix} y_t^{obs} \\ h_t^{obs} \\ \pi_t^{obs} \\ i_t^{obs} \\ w_t^{obs} \end{bmatrix} = \begin{bmatrix} y_t - y_{t-1} \\ h_t \\ \pi_t \\ i_t \\ \pi_t^w \end{bmatrix} \quad (3.1)$$

where all observables are demeaned prior to estimation and treated as stationary. Following Ratto et al. (2009), the estimation begins by locating the mode of the posterior distribution that maximizes the log-posterior function, which combines the information in the priors with the likelihood of the observed data. Next, a chain of 500,000 iterations of the Metropolis–Hastings algorithm is employed to obtain the full posterior distribution.

Prior distributions of the parameters. The prior distributions follow the literature on estimated DSGE models, such as Smets and Wouters (2007). Beginning with the exogenous disturbances, the standard deviations of the innovations are assigned inverse gamma priors, while the autoregressive coefficients are assigned beta priors.¹⁰ Turning to the structural parameters, the elasticities of inflation and output growth in the systematic component of monetary policy, the share of capital, the inverse of the Frisch elasticity, and the price and wage markups are assumed to follow normal distributions, whereas the parameters governing price and wage stickiness are given beta priors. The discount factor, expressed as $100 \cdot (1/\beta - 1) \equiv \tilde{\beta}$, follows the approach of Bilbiie et al. (2023) and is assigned a gamma prior. Finally, the newly introduced income elasticities, γ_s and γ_h , also follow gamma priors to ensure non-negativity, while the error-correction parameters, κ_s and κ_h , are assigned beta priors consistent with their support on the unit interval. In the estimation process, two parameters are fixed: the steady-state probabilities in matrix (2.2), set at $s = 0.9800$ and

¹⁰As noted in Subsection 2.5, during the estimation of the D-THANK and THANK models, lump-sum transfers are assumed to remain at their steady-state value of zero, reflecting their historically sporadic use.

$h = 0.9459$. These values imply a steady-state share of hand-to-mouth agents of 0.27 and are consistent with the average marginal propensity to consume out of transitory income shocks reported by Parker et al. (2013).

Parameter	Prior family	Prior mean	Prior s.d.	Post. mode	Post. mean	90% HPDI
<i>Preferences and technology</i>						
α	Normal	0.30	0.025	0.282	0.289	[0.248, 0.328]
φ	Normal	2.00	0.750	2.129	2.159	[1.000, 3.289]
$\tilde{\beta}$	Gamma	0.25	0.100	0.216	0.258	[0.094, 0.414]
$\mathcal{M}_p$	Normal	1.15	0.025	1.146	1.143	[1.101, 1.187]
$\mathcal{M}_w$	Normal	1.15	0.025	1.137	1.136	[1.093, 1.180]
θ_p	Beta	0.50	0.100	0.772	0.801	[0.698, 0.910]
θ_w	Beta	0.50	0.100	0.767	0.755	[0.679, 0.836]
<i>Monetary policy</i>						
ρ_i	Beta	0.50	0.200	0.814	0.826	[0.778, 0.876]
ϕ_π	Normal	1.50	0.250	2.033	1.979	[1.620, 2.339]
ϕ_y	Normal	0.125	0.050	0.257	0.246	[0.163, 0.328]
<i>Household composition (D-THANK only)</i>						
γ_s	Gamma	0.30	0.150	0.087	0.104	[0.024, 0.183]
γ_h	Gamma	0.30	0.150	0.634	0.827	[0.450, 1.202]
κ_s	Beta	0.15	0.070	0.203	0.222	[0.159, 0.300]
κ_h	Beta	0.15	0.070	0.099	0.121	[0.031, 0.209]
<i>Shock persistences</i>						
ρ_a	Beta	0.50	0.200	0.933	0.924	[0.884, 0.964]
ρ_{cp}	Beta	0.50	0.200	0.850	0.783	[0.650, 0.913]
ρ_w	Beta	0.50	0.200	0.176	0.186	[0.053, 0.312]
ρ_{pref}	Beta	0.50	0.200	0.923	0.911	[0.876, 0.947]
<i>Innovation standard deviations</i>						
σ_a	Inv. Gamma	0.10	2.000	0.443	0.452	[0.394, 0.511]
σ_{cp}	Inv. Gamma	0.10	2.000	0.047	0.055	[0.038, 0.070]
σ_w	Inv. Gamma	0.10	2.000	0.626	0.633	[0.515, 0.752]
σ_v	Inv. Gamma	0.10	2.000	0.121	0.121	[0.102, 0.140]
σ_{pref}	Inv. Gamma	0.10	2.000	0.121	0.155	[0.094, 0.217]

Table 1: Prior and posterior distributions of the D-THANK model

Note: Posterior mode, mean, and 90% highest posterior density interval (HPDI) are computed from 500,000 draws of the random-walk Metropolis–Hastings sampler with 20% burn-in. Sample: 1987Q1–2007Q4.

Table 1 reports the priors and posterior estimates for the D-THANK model. For comparison, the corresponding estimates for the THANK model are presented in Appendix D.2. Overall, the data are clearly informative, as evidenced by the pronounced updating from

prior to posterior distributions. At the same time, posterior estimates of the parameters common to both models are remarkably similar across specifications, indicating that introducing an endogenous household composition leaves the remaining structural features of the economy largely unchanged.

Identification of the switching parameters. Because the transition probabilities are not directly observed, a natural question is whether the parameters governing them can be separately identified.

Identification rests on the distinct roles the two probabilities play in the equilibrium system. The probability s_t enters the optimization problem of unconstrained households directly: because savers accumulate assets partly to insure against becoming hand-to-mouth, expected transitions into that state enter the Euler equation through a precautionary-saving motive. The probability h_t , by contrast, concerns households that make no intertemporal decision and therefore operates only through the law of motion for the aggregate share λ_t . As a result, alternative combinations of (s_t, h_t) that generate identical paths for that share nevertheless imply different saving behavior for unconstrained households, and hence different paths for the observables. Because such combinations are not observationally equivalent, the likelihood identifies γ_s and γ_h separately.¹¹

Whether the data in fact identify both margins—and, in particular, whether the asymmetry between γ_s and γ_h is a genuine feature of the data—is ultimately an empirical question. In principle, cyclical fluctuations in λ_t could arise from s_t alone, leaving the h_t margin as pure additional flexibility. To avoid imposing any asymmetry a priori, the priors on γ_s and γ_h are deliberately identical, so that any posterior divergence is driven entirely by the likelihood. The evidence indicates that both margins are operative. As Table 1 reports and the posterior densities in Appendix D.2 illustrate, the posteriors separate clearly, with γ_h

¹¹An alternative would be to construct transition probabilities directly from household panel data such as the PSID and include them among the observables. This approach, however, entails nontrivial measurement and modeling challenges. The PSID is biennial—a much lower frequency than the quarterly data used in the estimation—and mapping it into the model would require a substantially richer structure for household transitions. Addressing these issues lies beyond the scope of the present paper and is left for future research.

well above γ_s and bounded away from zero. The data therefore reject the restriction that compositional change operates through a single margin.

The estimated asymmetry also has a natural economic interpretation. The implied composite elasticities, $\tilde{\gamma}_h \equiv \lambda h \gamma_h \approx 0.21$ and $\tilde{\gamma}_s \equiv (1 - \lambda) s \gamma_s \approx 0.07$, indicate that fluctuations in disposable income affect the household distribution primarily through transitions of constrained households out of the hand-to-mouth state—a channel roughly three times as strong as the corresponding retention margin among unconstrained households. The underlying mechanism is intuitive. Households close to the borrowing constraint are highly sensitive to changes in disposable income, making transitions into unconstrained status more responsive to income fluctuations. Unconstrained households, by contrast, typically hold precautionary wealth buffers, making their transition probabilities far less responsive to comparable income fluctuations. This interpretation is consistent with the cyclical dynamics of the hand-to-mouth share documented in Bilbiie (2020). As an external validation, Appendix D.2 constructs an empirical counterpart to the hand-to-mouth share using data from the Consumer Expenditure Survey. Although no consensus high-frequency measure of this variable exists, the smoothed model-implied series closely tracks its empirical proxy, lending further support to the model’s implied household transition dynamics.

Empirical fit. It remains to assess whether the endogenous household distribution improves the model’s empirical fit. Because the D-THANK and THANK models are nested—differing only in whether $\gamma_s, \gamma_h, \kappa_s, \kappa_h$ are estimated or fixed at zero—the question can be answered directly by the marginal data density. This measure gives the probability the model assigns to the observed sample, obtained by integrating the likelihood over the prior. Higher values indicate better fit, and because the integration averages over the parameter space, additional parameters raise the measure only when the improvement in fit outweighs the added flexibility. The criterion therefore balances fit against parsimony automatically.

Table 2 reports the marginal data density, computed by the Laplace approximation and

by the modified harmonic mean estimator of Geweke (1999). For completeness, the table also reports the Akaike, Bayesian, and Hannan–Quinn information criteria (AIC, BIC, and HQC), for which lower values indicate the preferred model. The evidence uniformly favors the D-THANK model. Its log marginal data density is -224.57 under the Laplace approximation and -224.30 under the modified harmonic mean, against -231.79 and -231.55 for the THANK. A gap of this size is difficult to interpret on its own, since the marginal data density carries no natural scale; the log Bayes factor, given by this difference, provides one. On the classification of Kass and Raftery (1995), values above 5 constitute “very strong” evidence in favor of the more flexible model, and the differences here—7.22 and 7.25—clear that threshold comfortably. The information criteria point in the same direction, each favoring the D-THANK.

In sum, despite estimating four additional parameters, the D-THANK model achieves a substantially higher marginal data density and lower information criteria, indicating that allowing household composition to evolve endogenously improves the model’s empirical performance.

Criterion	D-THANK	THANK	Difference (D–T)
<i>Bayesian model comparison</i>			
Log marginal data density (Laplace)	-224.57	-231.79	$+7.22$
Log marginal data density (MHM)	-224.30	-231.55	$+7.25$
<i>Information criteria</i>			
AIC	388.22	408.80	-20.58
BIC	443.86	454.76	-10.90
HQC	410.57	427.26	-16.69

Table 2: Model comparison between the THANK and D-THANK models

Note: Both models are estimated on the same five observables over 1987Q1–2007Q4. Positive differences in log marginal data densities and negative differences in information criteria favor D-THANK. Log marginal data densities are computed by the Laplace approximation and the modified harmonic mean estimator; AIC, BIC, and HQC denote the Akaike, Bayesian, and Hannan–Quinn information criteria.

3.1 Impulse response analysis

Having shown that an endogenous household distribution improves the model’s fit, the next step is to examine how household composition shapes macroeconomic dynamics. The analysis proceeds in two steps. The first isolates the distribution channel by tracing the economy’s response to a pure shift in household composition. The second shows how allowing the distribution to adjust endogenously alters the transmission of a given structural disturbance relative to a model with a fixed share of constrained households. Together, these exercises establish the mechanism that informs the policy analysis that follows.

Consider first an exogenous increase in the share of constrained households, referred to as a “hand-to-mouth shock.” For illustrative purposes, Figure 2 augments the law of motion for λ_t with a one-period innovation u_t^λ , which is not part of the estimated model. The innovation is normalized to raise the hand-to-mouth share by one percent on impact.¹² Consistent with the evidence that low-income households consume less than high-income households, an increase in the hand-to-mouth share reduces aggregate demand and puts downward pressure on prices. The same shift in composition also lowers the average marginal rate of substitution between consumption and leisure, leading labor unions to set lower nominal wages. The resulting decline in inflation and economic activity induces the central bank to reduce the nominal interest rate. A shift in composition alone therefore propagates through the full set of macroeconomic aggregates.

The second exercise turns to a structural disturbance and asks how endogenous movements in the household distribution alter its transmission. Figure 3 compares the response to a cost-push shock in the D-THANK model (solid blue) and the THANK model (dashed orange), in which the hand-to-mouth share is held fixed.¹³ In both models, the shock raises firms’ desired markups and directly increases price inflation. At the same time, weaker out-

¹²Identical impulse responses could also be generated by a one-period innovation to either s_t or h_t , since one-period innovations to all three variables affect the system only through a common composite term. This holds for one-period shocks alone, as persistent innovations to s_t would additionally enter the Euler equation through the precautionary term $E_t \hat{s}_{t+1}$.

¹³For the remaining structural shocks, see Figures 11 to 13 in Appendix H.1.

put reduces labor demand and places downward pressure on wage inflation. In response, the central bank tightens monetary policy according to the Taylor rule. The two economies therefore differ not in the direction of their responses, but in their magnitude and persistence.

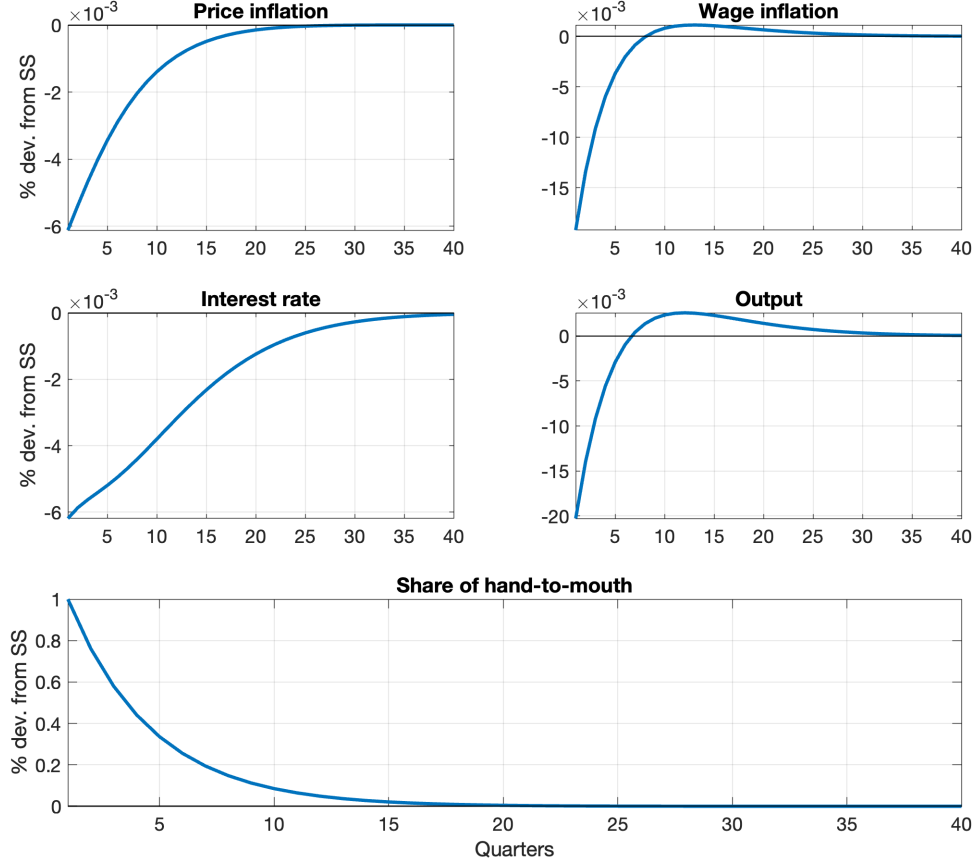


Figure 2: Impulse response functions to a hand-to-mouth shock

Note: Impulse responses are constructed at the posterior mode of the structural parameters. The hand-to-mouth shock is a one-period innovation to the law of motion for λ_t , normalized to raise the share of hand-to-mouth households by one percent on impact.

The source of these differences is the endogenous evolution of the household distribution. As disposable labor income declines, households become more likely to enter or remain in a financially constrained state, causing the hand-to-mouth share to rise. While this margin is absent in the THANK model, where the share remains fixed, in the D-THANK model it follows a pronounced hump-shaped response, peaking around 1.6 percent above

steady state after roughly eight quarters and gradually returning over the following quarters. This endogenous shift amplifies aggregate fluctuations through two channels. First, savers internalize the higher probability of becoming constrained in the future, increasing the value of liquidity and reducing current consumption through a stronger precautionary motive, a mechanism also identified in Bilbie et al. (2023). Second, the economy's composition shifts toward households with higher marginal propensities to consume, making aggregate demand more closely tied to current income and prolonging the downturn.

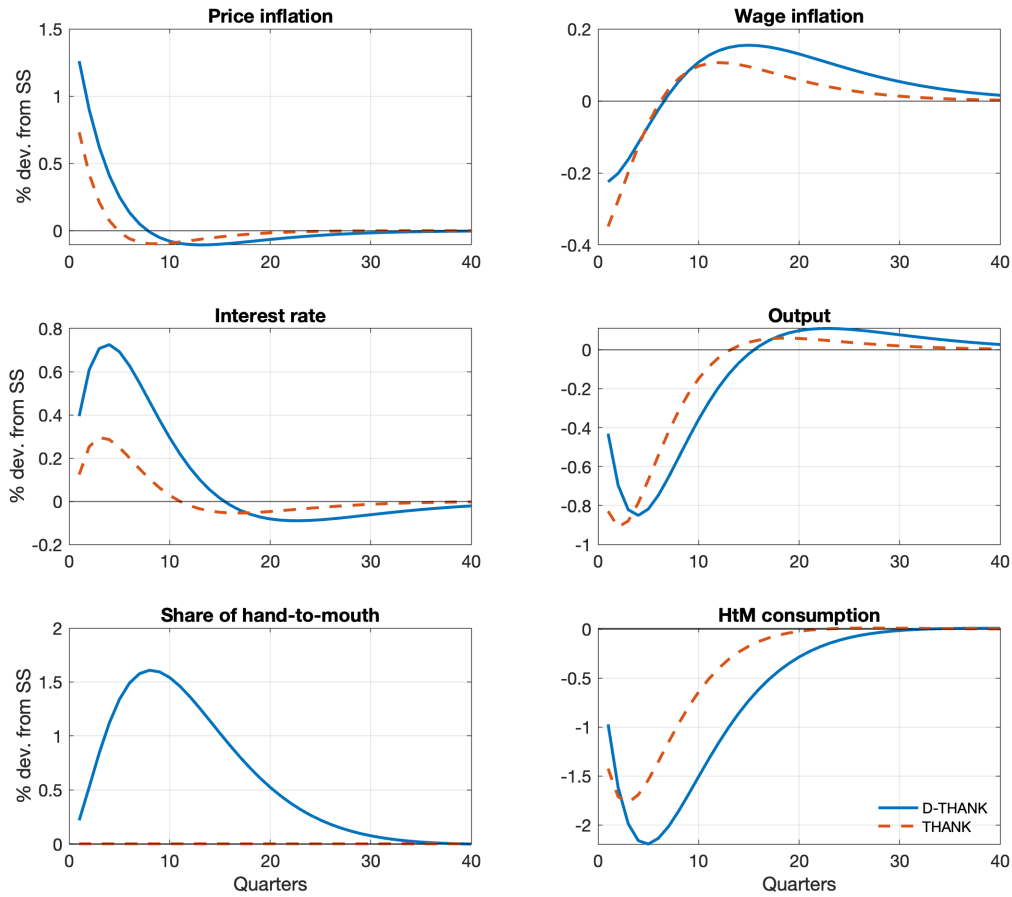


Figure 3: Impulse response functions to a cost-push shock

Note: Impulse responses are constructed at the posterior mode of the structural parameters. Both models are normalized to a 0.1 percent innovation to the cost-push process on impact. Solid blue lines refer to the D-THANK model and dashed orange lines to the THANK model.

Constrained households absorb the largest and most persistent income loss. Although

their disposable labor income contracts by less on impact than in the THANK model, about 1 percent against 1.5 percent, it continues to decline for five quarters, reaching a trough of approximately 2.3 percent, and recovers only after around thirty quarters—roughly ten quarters later than in the THANK model. The endogenous distribution thus does not merely amplify the shock but prolongs it, concentrating the cost on the households least able to smooth consumption. These features are central to the transfer analysis that follows, where the policy problem is to stabilize an economy in which downturns both disproportionately affect constrained households and increase their share.

4 Optimal fiscal transfers

This section derives a second-order approximation to the Ramsey planner’s welfare criterion and uses it to evaluate optimal and simple transfer rules under alternative financing regimes.

In the presence of household heterogeneity, the production subsidy that offsets monopolistic distortions in the standard New Keynesian model is no longer sufficient to implement the efficient steady state, because the distribution of consumption across household types introduces an additional welfare-relevant margin. Following the inverse-optimal-taxation literature (e.g., McKay and Wolf, 2023 and Heathcote and Tsujiyama, 2021), the Ramsey planner’s Pareto weights are therefore assigned residually—conditional on a given fiscal rule—to support an efficient steady state. Appendix B.2 provides further details.¹⁴

The remainder of the section is organized as follows. Subsection 4.1 derives the welfare loss function and characterizes its distributional component. Subsection 4.2 examines optimal and simple transfer rules under tax and debt financing and compares their welfare properties.

¹⁴Given the paper’s focus on the short-run role of fiscal transfers, this approach is preferable to holding Pareto weights fixed and restoring long-run efficiency through a combination of taxes and transfers that equalizes consumption across households.

4.1 Welfare function and inequality

The Ramsey planner aggregates the lifetime utilities of the two household groups according to

$$\mathcal{W} = \sum_{t=0}^{\infty} \beta^t \sum_{i=\{H,S\}} \varphi_i \left(\frac{(\omega_{it} C_t^A)^{1-\sigma}}{1-\sigma} - \frac{(N_t)^{1+\varphi}}{1+\varphi} \right), \quad (4.1)$$

where φ_i represents the Pareto weight for household group i and $\omega_{it} \equiv \tilde{C}_t^i / C_t^A$ its share of aggregate consumption, with $\tilde{C}_t^H \equiv \lambda_t C_t^H$ and $\tilde{C}_t^S \equiv (1 - \lambda_t) C_t^S$.

A second-order approximation around the efficient steady state yields the quadratic loss function

$$\mathcal{L} = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left[\chi_w (\pi_t^w)^2 + \chi_p (\pi_t^p)^2 + \chi_y (\hat{y}_t)^2 + \chi_S (\hat{\omega}_{S,t})^2 + \chi_H (\hat{\omega}_{H,t})^2 \right], \quad (4.2)$$

where $\chi_r > 0$, for $r \in \{w, p, y, S, H\}$, denotes the welfare weight on each component of the loss function (see Appendix E for the full derivation). The first three terms are the standard New Keynesian losses associated with wage inflation, price inflation, and inefficient fluctuations in aggregate output. The final two add a distributional component, penalizing deviations in the consumption shares of the two household groups:

$$\hat{\omega}_{H,t} = \bar{\omega}_H \left(\frac{1}{\lambda} \hat{\lambda}_t + \hat{c}_t^H - \hat{c}_t^A \right), \quad \hat{\omega}_{S,t} = \bar{\omega}_S \left(-\frac{1}{1-\lambda} \hat{\lambda}_t + \hat{c}_t^S - \hat{c}_t^A \right). \quad (4.3)$$

Accordingly, movements in these shares arise from two sources. The first is variation in type-specific consumption relative to aggregate consumption. The second stems from endogenous changes in household composition, which reallocate consumption across groups even when type-specific consumption is unchanged. This second channel is absent from conventional two-agent models with fixed household shares and introduces an additional margin through which heterogeneity shapes welfare and optimal policy.

Composition and inequality. The welfare loss admits an equivalent representation in terms of a single distributional target. Since household consumption shares sum to unity, the per-period loss can be written as

$$\mathcal{L}_t = \chi_w (\pi_t^w)^2 + \chi_p (\pi_t^p)^2 + \chi_y (\hat{y}_t)^2 + \chi_\omega (\hat{\omega}_{H,t})^2, \quad (4.4)$$

where $\chi_\omega \equiv (\bar{\omega}_H \bar{\omega}_S)^{-1}$.¹⁵ To make the forces underlying this distributional target explicit, substituting the aggregate consumption identity into (4.3) yields

$$\hat{\omega}_{H,t} = \Psi_c \hat{\lambda}_t + \Psi_g \hat{c}_t^G, \quad (4.5)$$

where $\hat{c}_t^G \equiv \hat{c}_t^H - \hat{c}_t^S$ is the consumption gap between hand-to-mouth households and savers, while the coefficients $\Psi_g \equiv \lambda(1 - \alpha)[\mathcal{M}_p - \lambda(1 - \alpha)]/\mathcal{M}_p^2$ and $\Psi_c \equiv \Psi_g/[\lambda(1 - \lambda)]$ are both positive over the admissible parameter space.

These two terms have distinct economic interpretations. The first, the *composition channel*, operates through changes in the fraction of financially constrained households. An increase in $\hat{\lambda}_t$ raises the hand-to-mouth consumption share by increasing the relative mass of constrained households. The second, the *relative-consumption channel*, operates through the consumption gap between household types. Holding household composition fixed, a larger $\hat{c}_t^G$ likewise raises the hand-to-mouth consumption share. As shown in Section 5, this share is countercyclical in both the model and the data.¹⁶

4.2 Optimal transfer rules

Fiscal transfers provide a direct instrument for stabilizing inequality over the business cycle. Implementing such transfers, however, requires specifying how the associated fiscal burden

¹⁵Equivalently, the distributional objective could be expressed in terms of the savers' consumption share. The hand-to-mouth share is used here because the analysis focuses on fiscal transfers targeted toward financially constrained households.

¹⁶Appendix F further characterizes the cyclical behavior of the consumption gap.

is financed. A natural benchmark is contemporaneous redistribution from savers to hand-to-mouth households, whereas in practice governments often rely on debt issuance to spread this burden over time. In a heterogeneous-agent environment, these two financing arrangements need not be equivalent, since the breakdown of Ricardian equivalence allows the timing of fiscal adjustment to affect both aggregate and distributional dynamics. I therefore consider optimal transfer policy under both regimes. To accommodate debt issuance, I relax the zero-liquidity assumption of Section 2 and extend the baseline D-THANK model with a fiscal rule in the spirit of Gali (2020). Appendix G derives the extended model.

The remainder of the subsection proceeds in three steps. First, it compares the macroeconomic and distributional effects of an exogenous transfer shock under tax and debt financing. Second, it characterizes the Ramsey-optimal transfer allocation under each regime. Third, it estimates simple, implementable transfer rules that approximate the Ramsey allocation and compares their welfare properties.

The effects of a transfer shock. Before turning to optimal policy, consider the transmission of an exogenous transfer shock under tax and debt financing. Figure 4 reports the corresponding impulse responses.

A transfer equivalent to 0.1% of steady-state output raises the disposable income and consumption of hand-to-mouth households, stimulating aggregate demand and output.¹⁷ The resulting increase in labor demand puts upward pressure on real wages and marginal costs, prompting the central bank to raise the nominal interest rate. Beyond these aggregate effects, the transfer also alters household composition. Higher disposable income increases transitions from the constrained to the saver state while reducing transitions in the opposite direction, thereby lowering the fraction of hand-to-mouth households. As shown in Figure 5, at longer horizons the composition channel dominates the relative-consumption channel, so

¹⁷Alternatively, transfers could be distributed universally across all households rather than targeted to hand-to-mouth households. Appendix H.3 compares the two designs under debt financing and shows that universal transfers are considerably less fiscally efficient, since providing the same support for constrained households would require an aggregate outlay approximately $1/\lambda \approx 3.7$ times larger, with correspondingly greater debt issuance and stronger inflationary pressures.

the consumption share of hand-to-mouth households falls.

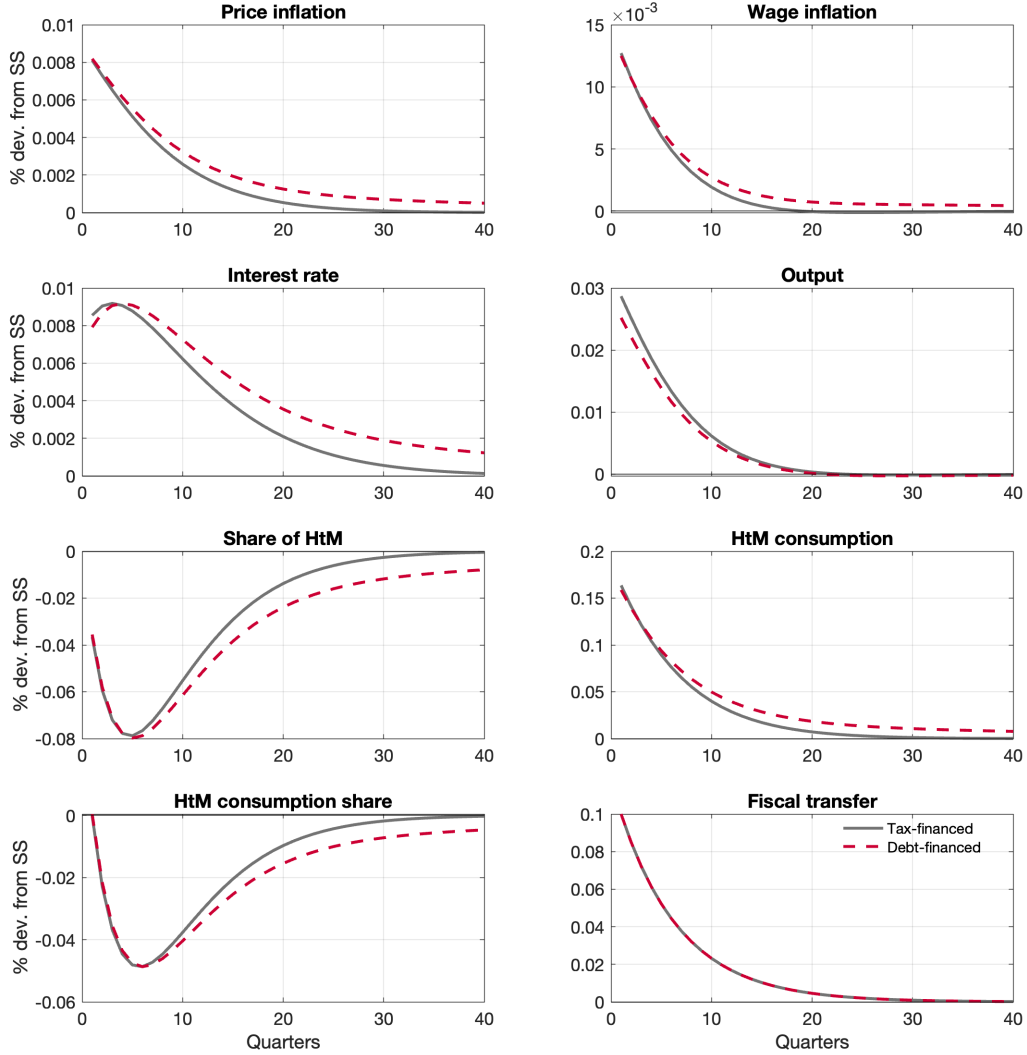


Figure 4: Impulse response functions to a transfer shock under alternative financing regimes
Note: Impulse responses are constructed at the posterior mode of the structural parameters. The exogenous transfer shock is normalized to deliver a transfer of 0.1% of steady-state output to hand-to-mouth households on impact, with a persistence parameter of 0.85. Solid gray lines refer to tax financing and dashed crimson lines to debt financing, as defined in Appendix G. The consumption share of hand-to-mouth households is constructed from equation (4.5).

Although the two financing regimes generate similar responses at short horizons, the dynamics diverge thereafter. Under debt financing, public debt accumulates following the transfer, peaking roughly two years after the shock before being gradually retired through the consolidation rule (see Figure 15 in the Appendix). The outstanding stock of debt interacts

with household transitions to prolong the expansion. In particular, savers who subsequently enter the constrained state retain their bond holdings and continue to receive interest income after the initial transfer has dissipated. At the same time, the prospect of a more persistent demand expansion, together with the associated monetary tightening, moderates the initial response of output under debt financing.

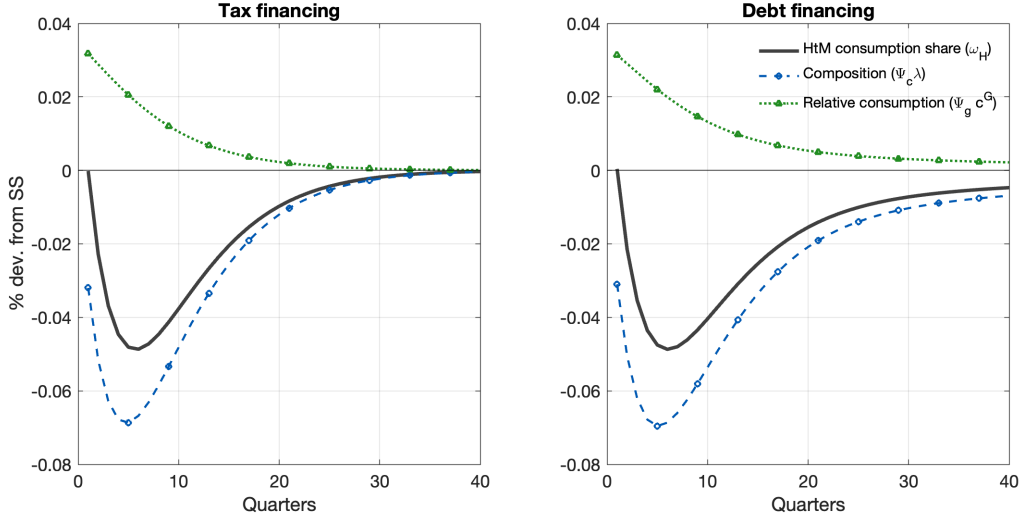


Figure 5: Decomposition of the response of the hand-to-mouth consumption share

Note: Each panel decomposes the response of the hand-to-mouth consumption share to the transfer shock of Figure 4 into the composition channel ($\Psi_c \lambda_t$) and the relative-consumption channel ($\Psi_g c_t^G$), following equation (4.5). Left panel: tax financing; right panel: debt financing.

Overall, while the effects of transfers are broadly similar across regimes on impact, the interaction between public debt and endogenous household transitions makes debt-financed transfers a more persistent form of redistribution.

Ramsey-optimal transfers. Having characterized the transmission of transfers under the two financing regimes, a natural next step is to determine how transfer policy should be optimally designed. As a normative benchmark, I first consider the Ramsey problem, which selects the path of fiscal transfers that maximizes welfare subject to the equilibrium conditions of the economy. The resulting allocation attains the highest welfare achievable through fiscal transfers and therefore provides the standard against which the simple transfer

rules considered below are evaluated.¹⁸

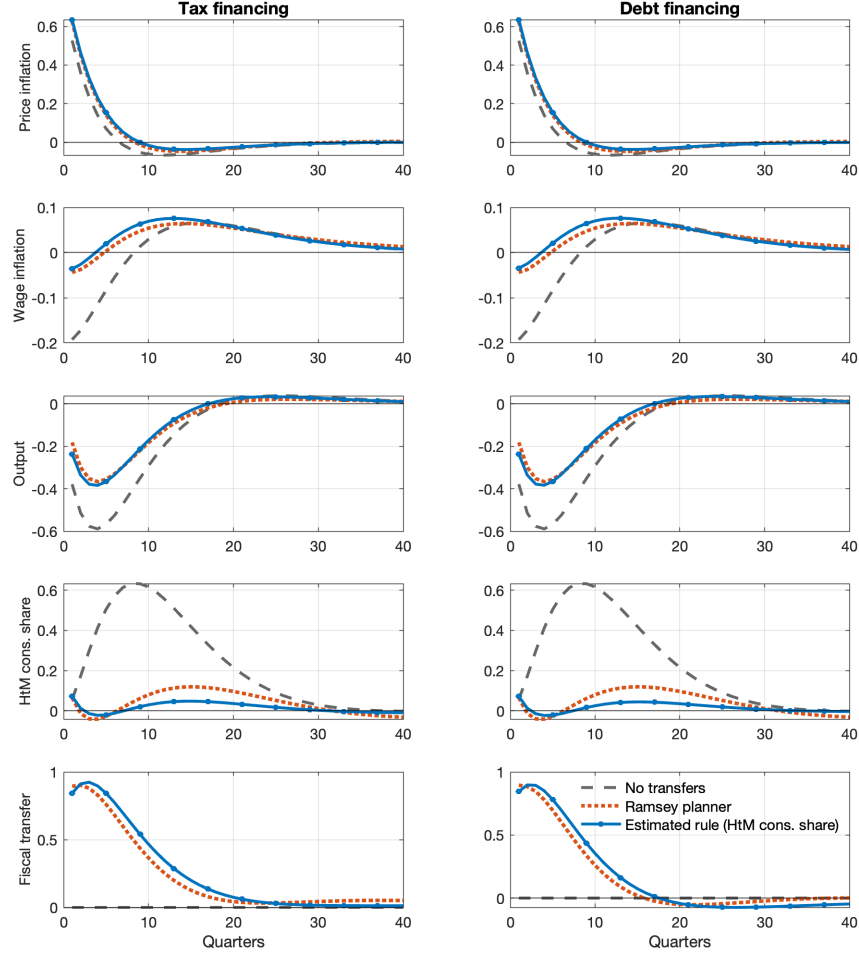


Figure 6: Responses to a cost-push shock under alternative transfer policies

Note: Impulse responses to a one-standard-deviation cost-push shock at the posterior mode. Left panels refer to tax financing and right panels to debt financing. Dashed gray lines report the equilibrium without transfers, dotted orange lines the Ramsey allocation, and solid blue lines the estimated rule responding to output and the consumption share of hand-to-mouth households.

To begin, Figure 6 illustrates the Ramsey response to a cost-push shock under the two financing regimes (dotted orange lines). The equilibrium without transfers (dashed gray lines) provides a useful reference for assessing how optimal transfer policy reshapes the adjustment to the shock.

As discussed in Section 3.1, absent transfers, the cost-push shock raises inflation and

¹⁸Throughout the optimal-policy analysis, the conduct of monetary policy is held fixed at the Taylor rule estimated in Section 3, so that the resulting transfer policy is optimal conditional on the observed behavior of the central bank.

contracts output. The accompanying decline in real income also has distributional consequences, increasing the fraction of constrained households and, with it, the share of aggregate consumption accounted for by these households. The Ramsey planner responds with a front-loaded transfer of approximately 1% of steady-state output under both financing regimes, roughly halving the output contraction on impact. Further stabilization of output would come at the cost of higher inflation, given the inflationary effects of fiscal transfers. The largest gains instead arise along the distributional margin, with optimal policy almost fully stabilizing the consumption share of hand-to-mouth households. Importantly, the Ramsey allocation is the same under the two financing regimes, but its implementation requires smaller transfers under debt financing, as debt issuance itself contributes to the fiscal stimulus through the bond windfall accruing to households that enter the constrained state. Responses of the remaining variables, together with the Ramsey responses to the other structural shocks, are reported in Appendix H.4.

Simple transfer rules. The Ramsey allocation establishes what transfer policy can achieve when the policymaker can condition on the full state of the economy. In practice, however, transfer policy must rely on simpler rules based on a limited set of observable indicators. The relevant question is therefore how much of the welfare gain under the Ramsey allocation can be retained by rules that are easier to implement.

To address this question, I estimate transfer rules directly from the Ramsey allocation by projecting optimal transfers on a small set of observable variables and then evaluating the welfare attained when each estimated rule is implemented in the model. I focus on rules that condition transfers on two economic indicators. The first is output, which summarizes aggregate cyclical conditions and captures the conventional stabilization motive for fiscal transfers. The second varies across specifications and is intended to provide information beyond aggregate activity, particularly about distributional conditions.

Because the relevant policy dynamics are not known *ex ante*, a flexible functional form is

desirable. To this end, the transfer rule is specified as an ARDL(1,1), allowing the estimation to determine both the persistence of transfers and the timing of their response to economic conditions:

$$\hat{t}_t^H = \rho_t \hat{t}_{t-1}^H + \phi_{y,0} \hat{y}_t + \phi_{y,1} \hat{y}_{t-1} + \phi_{x,0} \hat{x}_t + \phi_{x,1} \hat{x}_{t-1}, \quad (4.6)$$

where $\hat{x}_t$ denotes the second indicator. Candidate indicators include the consumption of hand-to-mouth households, the consumption gap, the real wage, hours worked, and the consumption share of hand-to-mouth households. I also consider a restricted specification in which transfers respond to output alone. The resulting rules are evaluated by their welfare distance from the Ramsey allocation, expressed as a consumption equivalent variation.

Policy rule	Tax financing		Debt financing	
	Welfare loss	C.E.V. (%)	Welfare loss	C.E.V. (%)
<i>Benchmarks</i>				
No transfers	152.05	12.41	152.05	12.41
Ramsey allocation	97.74	0.00	97.74	0.00
<i>Output-only rule</i>				
Output ($\hat{y}_t$)	115.76	3.96	114.37	3.65
<i>Additional indicators</i>				
HtM consumption ($\hat{c}_t^H$)	122.12	5.39	121.37	5.22
Consumption gap ($\hat{c}_t^G$)	123.13	5.62	121.53	5.26
Real wage ($\hat{\omega}_t$)	109.69	2.61	109.68	2.61
Hours worked ($\hat{n}_t$)	110.27	2.74	109.56	2.58
HtM cons. share ($\hat{\omega}_{H,t}$)	103.73	1.30	103.73	1.30

Table 3: Welfare properties of estimated transfer rules

Note: For each candidate indicator, the coefficients of equation (4.6) are estimated by ordinary least squares on the paths of transfers and indicators simulated under the Ramsey policy. Welfare losses are computed by simulating the economy under each estimated rule and evaluating the loss function (4.2). The column C.E.V. reports the welfare distance of each policy from the Ramsey allocation, expressed as a consumption equivalent variation in percent, so that smaller values indicate rules closer to the optimal policy. Details on the derivation of consumption equivalent variations are provided in Appendix E.

Welfare analysis. Table 3 shows that simple transfer rules retain a substantial share of the welfare gains attainable under optimal policy. The output-only rule already delivers a sizable improvement over the no-transfer allocation, with the welfare distance from the Ramsey allocation falling from 12.41% to about 4% of permanent consumption under both

financing regimes. The remaining gap, however, points to an important role for information beyond aggregate activity, and the choice of the second indicator matters considerably for the performance of the rule. In particular, among the candidates considered, the consumption share of hand-to-mouth households performs best, reducing the welfare distance from the Ramsey allocation to 1.30%. The estimated rule prescribes countercyclical transfers, increasing support when output falls and when the consumption share of hand-to-mouth households rises. Appendix H.5 illustrates its fit to the Ramsey allocation and reports the estimated coefficients.

The performance of the consumption-share rule also highlights an important distinction between the D-THANK framework and its THANK counterpart. With a fixed fraction of hand-to-mouth households, movements in the consumption share are driven entirely by the consumption gap ($\hat{c}_t^G$) between household types, as follows from equation (4.5). In the D-THANK economy, by contrast, the consumption share also reflects endogenous changes in household composition. The superior performance of the consumption-share rule relative to the consumption-gap rule therefore suggests that variation in the fraction of constrained households contains information relevant for optimal transfer policy beyond that conveyed by relative consumption alone.

The volatility decomposition clarifies the source of these gains. Figure 7 reports the change in the standard deviation of each component of the loss function under the estimated rules relative to the no-transfer allocation. While the rules achieve broadly similar stabilization of output at the cost of moderately higher inflation volatility, their distributional effects differ markedly. In particular, the consumption-share rule reduces the volatility of household consumption shares by nearly 80%, substantially more than the alternative specifications.

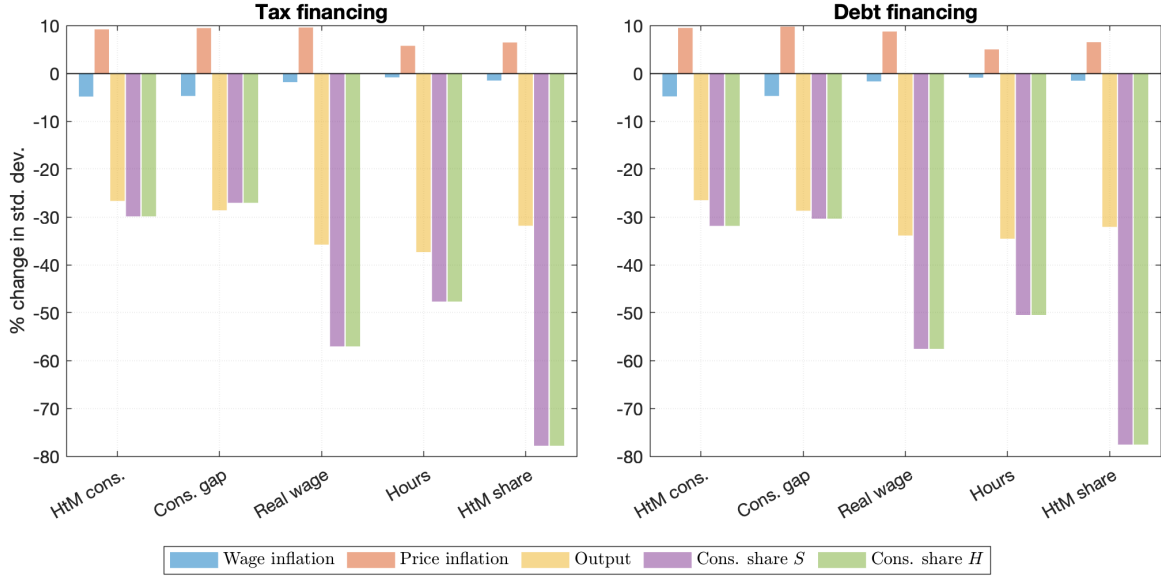


Figure 7: Welfare decomposition under the estimated transfer rules

Note: Percentage change in the standard deviation of each component of the loss function (4.2) under the estimated rules of Table 3, relative to the no-transfer allocation. The left panel refers to tax financing and the right panel to debt financing. Each group of bars corresponds to the rule combining output with the indicated additional indicator.

The main results are also largely invariant to the financing regime. Although debt financing is associated with lower welfare losses for several of the estimated rules, the differences relative to tax financing become negligible for the best-performing specifications. In particular, the consumption-share rule attains the same welfare under both regimes. Beyond welfare, the same rule also closely tracks the dynamics of the Ramsey allocation, including the near-complete stabilization of the consumption share of hand-to-mouth households shown in Figure 6.

5 COVID-19 counterfactual analysis

The results developed in Sections 3 and 4 are ultimately illustrated through a counterfactual analysis of the COVID-19 recession. The exercise quantifies the extent to which fiscal stimulus payments mitigated household income losses during the pandemic, and examines how the magnitude and timing of the fiscal response would have differed had transfers fol-

lowed the consumption-share rule from Section 4. Appendix I details the construction of the counterfactual.

The scale and timing of fiscal transfers. Did the COVID-19 transfer payments provide the appropriate amount of fiscal support at the right time? The counterfactual indicates that fiscal intervention played an important role in stabilizing household income during the pandemic recession, but also points to substantial differences between the realized response and the policy implied by the consumption-share rule. In the absence of transfers, household income would have fallen approximately 6% below its pre-pandemic level at the trough of the recession (black lines with circles in the right panel of Figure 8).¹⁹ The realized fiscal response more than offset this decline, with the first round of payments—roughly 10 percentage points of GDP—raising household income to almost 7% above its pre-pandemic level, and a second large round of payments following as the economy recovered (orange lines with triangles).

The consumption-share rule implies a considerably smaller response (blue lines with squares). Fiscal support rises by approximately 3 percentage points of GDP at the onset of the recession, reducing the decline in household income at the trough by roughly 3 percentage points.²⁰ The two policy paths diverge further during the recovery. Under the rule, transfers decline as economic conditions improve and fall below 1 percentage point of GDP. Realized transfers instead increased again with a second large round of payments, raising household income to almost 12% above its pre-pandemic level. The counterfactual thus suggests that prompt fiscal support at the onset of the recession was warranted, but that substantial stabilization of household income could have been achieved with a considerably smaller and more state-contingent response.

¹⁹As a complementary exercise, Appendix I also considers a counterfactual in which no fiscal transfers are implemented and uses the D-THANK model to distinguish the role of households' self-insurance behavior from the effects of endogenous changes in household composition.

²⁰The magnitude is comparable to Bayer et al. (2023), who find that fiscal stimulus during the COVID-19 pandemic reduced the decline in output by approximately two percentage points at the trough.

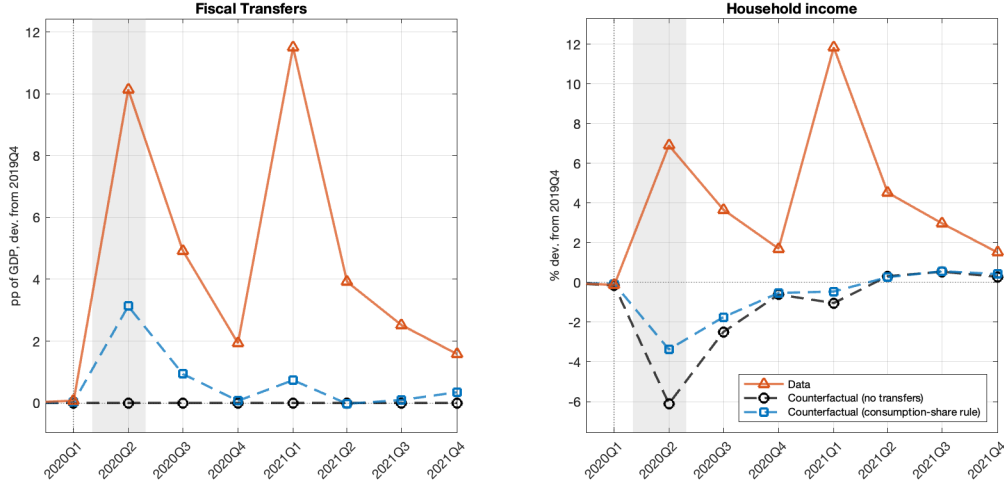


Figure 8: COVID-19 counterfactual: fiscal transfers and household income

Note: The counterfactual scenarios are constructed by feeding the smoothed productivity and preference shocks, recovered from the observed dynamics of real personal income excluding transfers and per-capita hours over 2017Q1–2022Q1, into the D-THANK model evaluated at the posterior mode, as detailed in Appendix I. The left panel reports aggregate fiscal transfers in percentage points of GDP and the right panel reports household income in percent, both as deviations from 2019Q4. Black lines refer to the counterfactual without transfers, blue lines to the consumption-share rule, and orange lines to the data, where household income corresponds to real personal income including transfers. Shaded areas denote NBER recessions.

6 Conclusions

This paper studies the design of simple fiscal transfer rules targeted to financially constrained, or hand-to-mouth, households. The analysis uses an estimated two-agent New Keynesian model in which households transition endogenously between constrained and unconstrained states. In this framework, aggregate shocks affect not only the consumption of each household type but also the relative size of the constrained population.

Two main results emerge. First, allowing households to transition endogenously between constrained and unconstrained states improves the model’s empirical fit relative to a specification with fixed household shares. Second, conditioning transfers on output and the hand-to-mouth share of aggregate consumption most closely approximates the Ramsey allocation among the rules considered, under both tax and debt financing. The hand-to-mouth share of aggregate consumption provides a particularly informative policy indicator, reflecting both consumption differences across household types and variation in the fraction of constrained households.

A counterfactual analysis of the COVID-19 recession further illustrates that, while prompt fiscal support at the onset of the recession was warranted, a considerably smaller and more state-contingent response—approximately 3 percentage points of GDP, compared with roughly 10 and 12 percentage points under the first and second rounds of realized payments—could have achieved substantial stabilization of household income at a significantly lower fiscal cost.

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Appendix

A Nonlinear equilibrium conditions

This appendix collects the complete set of nonlinear equilibrium conditions of the D-THANK model derived in Section 2, grouped by block.

A.1 Households

$$(C_t^S)^{-1} = \beta E_t \left[\frac{1 + i_t}{1 + \pi_{t+1}} (s_{t+1}(C_{t+1}^S)^{-1} + (1 - s_{t+1})(C_{t+1}^H)^{-1}) \right] \quad (\text{A.1})$$

$$C_t^H = \frac{W_t}{P_t} N_t + T_t^H \quad (\text{A.2})$$

$$Y_t^D = C_t^H \quad (\text{A.3})$$

A.2 Labor unions

$$W_t^* = \mathcal{M}_w \frac{E_t \sum_{k=0}^{\infty} (\beta \theta_w)^k N_{t+k}^{1+\varphi}(z)}{E_t \sum_{k=0}^{\infty} (\beta \theta_w)^k \left(\frac{\lambda_{t+k}}{C_{t+k}^H} + \frac{1-\lambda_{t+k}}{C_{t+k}^S} \right) \frac{N_{t+k}(z)}{P_{t+k}}} \quad (\text{A.4})$$

$$W_t^{1-\varepsilon_w} = \theta_w W_{t-1}^{1-\varepsilon_w} + (1 - \theta_w)(W_t^*)^{1-\varepsilon_w} \quad (\text{A.5})$$

A.3 Production

$$MC_t = \frac{W_t}{P_t(1 - \alpha)A_t N_t^{-\alpha}} \quad (\text{A.6})$$

$$\frac{P_t^*}{P_{t-1}} = \mathcal{M}_p \frac{E_t \sum_{k=0}^{\infty} \theta_p^k \Lambda_{t,t+k}^S Y_{t+k}(j) [MC_{t+k}(j) \Pi_{t-1,t+k}]}{E_t \sum_{k=0}^{\infty} \theta_p^k \Lambda_{t,t+k}^S Y_{t+k}(j)} \quad (\text{A.7})$$

$$P_t^{1-\varepsilon_p} = \theta_p P_{t-1}^{1-\varepsilon_p} + (1 - \theta_p)(P_t^*)^{1-\varepsilon_p} \quad (\text{A.8})$$

$$1 + \pi_t = \frac{P_t}{P_{t-1}} \quad (\text{A.9})$$

$$\log A_t = \rho_a \log A_{t-1} + u_t^a \quad (\text{A.10})$$

A.4 Policy

$$i_t = \rho_i i_{t-1} + (1 - \rho_i) [\phi_\pi \pi_t + \phi_y (\hat{y}_t - \hat{y}_{t-1})] + u_t^v \quad (\text{A.11})$$

$$T_t^H \lambda_t + T_t^S (1 - \lambda_t) = 0 \quad (\text{A.12})$$

A.5 Heterogeneity

$$\lambda_t = (1 - \lambda_{t-1})(1 - s_t) + \lambda_{t-1} h_t \quad (\text{A.13})$$

$$s_t = s \left(\frac{Y_t^D}{Y^D} \right)^{\gamma_s} e^{\kappa_s (\lambda_{t-1} - \lambda)} \quad (\text{A.14})$$

$$h_t = h \left(\frac{Y_t^D}{Y^D} \right)^{-\gamma_h} e^{-\kappa_h (\lambda_{t-1} - \lambda)} \quad (\text{A.15})$$

A.6 Aggregation and market clearing

$$Y_t v_t^p = A_t N_t^{1-\alpha} \quad (\text{A.16})$$

$$v_t^p = (1 - \theta_p) \left(\frac{P_t^*}{P_t} \right)^{-\varepsilon_p} + \theta_p \left(\frac{P_{t-1}}{P_t} \right)^{-\varepsilon_p} v_{t-1}^p \quad (\text{A.17})$$

$$N_t = N_t^D v_t^w \quad (\text{A.18})$$

$$v_t^w = (1 - \theta_w) \left(\frac{W_t^*}{W_t} \right)^{-\varepsilon_w} + \theta_w \left(\frac{W_{t-1}}{W_t} \right)^{-\varepsilon_w} v_{t-1}^w \quad (\text{A.19})$$

$$C_t^A = \lambda_t C_t^H + (1 - \lambda_t) C_t^S \quad (\text{A.20})$$

$$Y_t = C_t^A \quad (\text{A.21})$$

where the dispersion terms $v_t^p = \int_0^1 (P_t(j)/P_t)^{-\varepsilon_p} dj$ and $v_t^w = \int_0^1 (W_t(z)/W_t)^{-\varepsilon_w} dz$ can equivalently be rewritten in the recursive forms of Equations (A.17) and (A.19), which depend only on aggregate variables.

The system comprises 21 equations in the 21 endogenous variables

$$\{C_t^S, C_t^H, C_t^A, Y_t, Y_t^D, N_t, N_t^D, W_t, W_t^*, P_t, P_t^*, MC_t, \pi_t, i_t, \lambda_t, s_t, h_t, v_t^p, v_t^w, T_t^H, T_t^S\},$$

with A_t determined by the exogenous productivity process in Equation (A.10). In the baseline estimation, transfers remain at their zero steady-state value ($T_t^H = 0$), so that the fiscal balance in Equation (A.12) and the two transfer variables drop out, leaving 19 equations in 19 unknowns. In the optimal-policy analysis of Section 4, the transfer rule restores the closing condition.

B Steady states

B.1 Model estimation

The model is estimated around an inefficient steady state, as in Bilbiie et al. (2023). In steady state all variables are constant, expectations hold trivially, and the productivity process (A.10) is at its mean, $A = 1$. The transition probabilities are at their steady-state values s and h , since $Y_t^D = Y^D$ and $\lambda_{t-1} = \lambda$ in Equations (A.14)–(A.15), and transfers are zero, so the hand-to-mouth budget (A.2) reduces to $C^H = \frac{w}{p}N$. Given the parameters $\{\alpha, \beta, \varphi, \lambda, s, \mathcal{M}_p, \mathcal{M}_w\}$, the steady state is computed sequentially.

First, evaluating the marginal-cost condition (A.6) together with the production function (A.16) at the steady state, hand-to-mouth consumption is a fixed fraction of output,

$$C^H = \frac{1 - \alpha}{\mathcal{M}_p} Y, \quad (\text{B.1})$$

so constrained households consume a share $(1 - \alpha)/\mathcal{M}_p < 1$ of output.

Second, combining the resource constraint (A.21), the aggregate-consumption identity (A.20), and (B.1) gives saver consumption,

$$C^S = \frac{\mathcal{M}_p - \lambda(1 - \alpha)}{(1 - \lambda)\mathcal{M}_p} Y. \quad (\text{B.2})$$

Since $\mathcal{M}_p > 1 - \alpha$, savers consume more than hand-to-mouth households, $C^S > C^H$.

Third, define

$$\Omega \equiv \frac{\lambda \mathcal{M}_p}{1 - \alpha} + \frac{(1 - \lambda)^2 \mathcal{M}_p}{\mathcal{M}_p - \lambda(1 - \alpha)}. \quad (\text{B.3})$$

Combining the steady-state wage-setting condition (A.4) with the marginal-cost condition (A.6) and the consumption expressions (B.1)–(B.2) yields steady-state hours,

$$N = \left(\frac{(1 - \alpha) \Omega}{\mathcal{M}_p \mathcal{M}_w} \right)^{\frac{1}{1 + \varphi}}. \quad (\text{B.4})$$

The remaining quantities follow from (B.4) through the production function (A.16), the marginal-cost condition (A.6), and the resource constraint (A.21),

$$Y = N^{1-\alpha}, \quad \frac{W}{P} = \frac{1-\alpha}{\mathcal{M}_p} N^{-\alpha}, \quad C^A = Y. \quad (\text{B.5})$$

Finally, evaluating the saver Euler equation (A.1) at the steady state pins down the nominal interest rate,

$$1+i = \frac{1}{\beta \left[s + (1-s) \frac{C^S}{C^H} \right]}. \quad (\text{B.6})$$

Because $C^S/C^H > 1$, it follows that $\beta(1+i) < 1$: the steady-state real rate lies below the rate of time preference, reflecting the precautionary saving motive induced by the risk of transitioning to the constrained state.

B.2 Optimal policy analysis

For the optimal-policy and welfare analysis, the economy is assumed to operate at an efficient steady state. Efficiency is achieved through a production subsidy τ on firms' labor demand, financed by a lump-sum tax on firms.

The subsidy enters the steady-state labor-demand condition, modifying the marginal-cost relation (A.6) to

$$\frac{W}{P}(1-\tau) = \frac{1-\alpha}{\mathcal{M}_p} N^{-\alpha}. \quad (\text{B.7})$$

Combining this with the steady-state wage-setting condition (A.4),

$$\frac{W}{P} = \mathcal{M}_w N^\varphi \left(\frac{\lambda}{C^H} + \frac{1-\lambda}{C^S} \right)^{-1}, \quad (\text{B.8})$$

the subsidy that equates the two markups, and thereby restores the efficiency condition $\bar{u}_c(1-\alpha)N^{-\alpha} = N^\varphi$, is

$$\tau = 1 - \frac{1}{\mathcal{M}_p \mathcal{M}_w}. \quad (\text{B.9})$$

At this efficient steady state, dividends remain strictly positive,

$$D = Y [1 - (1 - \alpha)\mathcal{M}_w] > 0, \quad (\text{B.10})$$

so that savers, who hold the entire equity stream, consume more than hand-to-mouth households. The planner's Pareto weights are selected so that the average marginal utility of consumption internalized by the labor unions coincides with the marginal utility of aggregate consumption used by the planner. The full second-order welfare derivation, including the resulting loss function, is presented in Appendix E.

C Log-linear model

C.1 Log-linear equilibrium conditions in the D-THANK model

No.	Name	Equation
1	Wage markup	$\hat{\mu}_t^w = \hat{\omega}_t - \eta_H \hat{c}_t^H - \eta_S \hat{c}_t^S - \varphi \hat{n}_t$
2	Wage Phillips curve	$\pi_t^w = \beta E_t[\pi_{t+1}^w] - \psi_w \hat{\mu}_t^w - \psi_{w\lambda} \hat{\lambda}_t + \varepsilon_t^w$
3	Change in real wage	$\Delta \hat{\omega}_t = \pi_t^w - \pi_t^p$
4	Euler, savers	$\hat{c}_t^S = \gamma_S E_t \hat{c}_{t+1}^S + \gamma_H E_t \hat{c}_{t+1}^H - \left(\frac{1}{1+i^*} \hat{i}_t - E_t \pi_{t+1} \right) - \left(\frac{\gamma_S}{s} - \frac{\gamma_H}{1-s} \right) E_t \hat{s}_{t+1} + \varepsilon_t^{pref}$
5	Disposable income	$\hat{y}_t^D = \hat{\omega}_t + \hat{n}_t$
6	HtM consumption	$\hat{c}_t^H = \hat{y}_t^D$
7	Price markup	$\hat{\mu}_t^p = -\hat{\omega}_t + \hat{y}_t - \hat{n}_t$
8	Price Phillips curve	$\pi_t^p = \beta E_t[\pi_{t+1}^p] - \psi_p \hat{\mu}_t^p + \varepsilon_t^{cp}$
9	Production function	$\hat{y}_t = \hat{a}_t + (1 - \alpha) \hat{n}_t$
10	Aggregate consumption	$\hat{c}_t^A = -\psi_\lambda \hat{\lambda}_t + \psi_H \hat{c}_t^H + \psi_S \hat{c}_t^S$
11	Resource constraint	$\hat{y}_t = \hat{c}_t^A$
12	Taylor rule	$\hat{i}_t = \rho_i \hat{i}_{t-1} + (1 - \rho_i)(\phi_\pi \pi_t^p + \phi_y(\hat{y}_t - \hat{y}_{t-1})) + \sigma_v u_t^v$
13	Household composition	$\hat{\lambda}_t = (s + h - 1) \hat{\lambda}_{t-1} - (1 - \lambda) \hat{s}_t + \lambda \hat{h}_t$
14	Process for s	$\hat{s}_t = s \gamma_s \hat{y}_t^D + s \kappa_s \hat{\lambda}_{t-1}$
15	Process for h	$\hat{h}_t = -h \gamma_h \hat{y}_t^D - h \kappa_h \hat{\lambda}_{t-1}$
16	TFP shock	$\hat{a}_t = \rho_a \hat{a}_{t-1} + \sigma_a u_t^a$
17	Cost-push shock	$\varepsilon_t^{cp} = \rho_{cp} \varepsilon_{t-1}^{cp} + \sigma_{cp} u_t^{cp}$
18	Wage markup shock	$\varepsilon_t^w = \rho_w \varepsilon_{t-1}^w + \sigma_w u_t^w$
19	Preference shock	$\varepsilon_t^{pref} = \rho_{pref} \varepsilon_{t-1}^{pref} + \sigma_{pref} u_t^{pref}$

Table 4: Log-linear equilibrium conditions: D-THANK model.

Note: Beyond the productivity shock u_t^a and the monetary policy shock u_t^v already introduced in Section 2, the estimated model includes a cost-push shock ε_t^{cp} in the price Phillips curve, a wage-markup shock ε_t^w in the wage Phillips curve, and a preference shock ε_t^{pref} in the saver Euler equation. These additional disturbances are standard in estimated New Keynesian models and are included so that the number of structural shocks matches the observable series used in estimation.

D Data and estimation results

D.1 Data

Figure 1. The household data are the public-use microdata of the Consumer Expenditure Survey, Interview component, covering 1994Q1 onward (<https://www.bls.gov/cex/pumd.htm>). The unit of observation is the consumer unit–interview, weighted by the CEX population weight (`finlwt21`), with the top and bottom 1% of total expenditure trimmed within each quarter. Households are restricted to those with positive income and expenditure. A consumer unit is classified as hand-to-mouth in a given quarter if its before-tax family income (`fincbtax`), measured over the preceding twelve months, falls below $x\%$ of the federal poverty guideline for its family size and year, obtained from the HHS Office of the Assistant Secretary for Planning and Evaluation (ASPE, <https://aspe.hhs.gov/topics/poverty-economic-mobility/poverty-guidelines>). The baseline threshold is $x = 200$. This anchor is selected on two independent grounds. First, internal consistency with the model, since at $x = 200$ the sample-average share is 0.286, close to the model’s steady-state hand-to-mouth share of 0.27. Second, external evidence from Aguiar et al. (2024), whose reported mean incomes for wealth-defined hand-to-mouth households translate into poverty-guideline multiples that bracket this value.

Estimation. The model is estimated using quarterly data from 1987Q1 to 2007Q4. The sample begins with the start of the Greenspan chairmanship and ends before the onset of the Great Financial Crisis to avoid the period associated with the effective lower bound. The observables are per-capita output growth, GDP deflator inflation, the federal funds rate, per-capita hours worked, and nominal wage growth. All raw data are obtained from the Federal Reserve Economic Data (FRED). Per-capita variables are constructed using the civilian noninstitutional population aged 16 and over (`CNP160V`). Because this series exhibits discrete level shifts at census revision dates, it is first converted to quarterly averages and

then smoothed using a Hodrick–Prescott filter with smoothing parameter $\lambda = 1600$. Real GDP (GDPC1) and nonfarm business hours (HOANBS) are divided by this estimated population trend; per-capita output then enters in growth rates, while per-capita hours enter in levels.

Simulations. The COVID-19 counterfactual of Section 5 uses quarterly U.S. data over 2017Q1 to 2022Q1. The two observables entering the smoother are the per-capita growth rate of real personal income excluding transfers (W875RX1) and the level of per-capita hours (HOANBS), both demeaned over the window, with population measured by the civilian non-institutional population (CNP160V). Realized fiscal transfers, used only as an overlay in the figures, are measured by personal current transfer receipts (A063RC1Q027SBEA) as a share of trend GDP, and realized household income by real personal income including transfers (RPI). All series are expressed as deviations from 2019Q4, the last pre-pandemic quarter. The construction of the counterfactual is described in Appendix I.

Tables 5 and 6 provide detailed descriptions of the series used in the estimation and counterfactual exercises.

Label	Description	Frequency	Source
<i>Estimation observables</i>			
GDPC1	Real Gross Domestic Product	Q	FRED
GDPDEF	Gross Domestic Product: Implicit Price Deflator	Q	FRED
FEDFUNDS	Effective Federal Funds Rate	M	FRED
HOANBS	Nonfarm Business Sector: Hours Worked	Q	FRED
COMPNFB	Nonfarm Business Sector: Compensation Per Hour	Q	FRED
CNP160V	Civilian Noninstitutional Population, 16 and Over	M	FRED
<i>Additional series for the counterfactual</i>			
W875RX1	Real Personal Income Excl. Current Transfer Receipts	M	FRED
A063RC1Q027SBEA	Personal Current Transfer Receipts	Q	FRED
RPI	Real Personal Income	M	FRED

Table 5: Data series description and sources

Label	Construction
<i>Estimation observables</i>	
Y_t	$\ln (GDPC1_t / Pop_t^{trend})$
P_t	$\ln (GDPDEF_t)$
H_t	$\ln (HOANBS_t / Pop_t^{trend})$
W_t	$\ln (COMPNEFB_t)$
y_t^{obs}	$100 \times (Y_t - Y_{t-1}) - mean(100 \times (Y_t - Y_{t-1}))$
π_t^{obs}	$100 \times (P_t - P_{t-1}) - mean(100 \times (P_t - P_{t-1}))$
i_t^{obs}	$100 \times \ln [1 + \frac{FEDFUNDS_t}{400}] - mean(100 \times \ln [1 + \frac{FEDFUNDS_t}{400}])$
h_t^{obs}	$100 \times [H_t] - mean(100 \times [H_t])$
w_t^{obs}	$100 \times (W_t - W_{t-1}) - mean(100 \times (W_t - W_{t-1}))$
<i>Additional series for the counterfactual</i>	
INC_t	$\ln (W875RX1_t / Pop_t^{trend})$
RPI_t^{pc}	$\ln (RPI_t / Pop_t^{trend})$
$GDP_t^{\$,trend}$	$HP_{1600}(GDPC1_t \times GDPDEF_t / 100)$
inc_t^{obs}	$100 \times (INC_t - INC_{t-1}) - mean(100 \times (INC_t - INC_{t-1}))$
t_t^H	$100 \times A063RC1Q027SBEA_t / GDP_t^{\$,trend}$ (rel. to 2019Q4)
rpi_t^{obs}	$100 \times (RPI_t^{pc} - RPI_{t-1}^{pc}) - mean(100 \times (RPI_t^{pc} - RPI_{t-1}^{pc}))$

Table 6: Data series in the model

Note: Pop_t^{trend} is the HP(1600) trend of the quarterly-averaged population CNP160V. The five estimation observables enter the likelihood over 1987Q1–2007Q4. In the counterfactual (2017Q1–2022Q1), the Kalman smoother is fed only inc_t^{obs} and h_t^{obs} ; the transfer share t_t^H and rpi_t^{obs} serve as figure overlays.

The hand-to-mouth consumption share. The model identifies the consumption share of hand-to-mouth households as the central distributional target of the policy analysis. To examine its empirical counterpart, I construct a corresponding measure from the Consumer Expenditure Survey (CEX) by combining the household classification introduced above with household-level consumption expenditures. For a given threshold $x\%$ of the federal poverty guideline, the share is defined as

$$\omega_{H,t}(x) = \frac{\sum_{i \in \text{HtM}_t^x} w_{it} e_{it}}{\sum_i w_{it} e_{it}}, \quad (\text{D.1})$$

where HtM_t^x denotes the set of households classified as hand-to-mouth, w_{it} is the population weight of household i , and e_{it} denotes household consumption expenditure. The measure therefore captures the fraction of aggregate consumption accounted for by households clas-

sified as hand-to-mouth at each point in time.

To align the empirical measure with the consumption concept in the model, I define e_{it} as expenditures on nondurable goods and services. These expenditures are constructed from the CEX monthly expenditure files, which record spending at the Universal Classification Code (UCC) level for the reference months associated with each interview. Individual UCCs are classified as nondurables, services, or excluded according to the mapping reported in Table 7. Because expenditures exhibit pronounced seasonal variation, the numerator and denominator of equation (D.1) are seasonally adjusted separately using X-13ARIMA-SEATS before computing the ratio.

The resulting measure also provides a useful comparison with the steady-state consumption share implied by the model. Using the broader measure of total household expenditures, the average share is 14.4%, close to the model-implied value of $\psi_H = \lambda(1 - \alpha)/\mathcal{M}_p \approx 0.15$ at the posterior mode. Under the preferred consumption measure of nondurable goods and services, the corresponding average is 21.4%.

Figure 9 reports the behavior of the measure during the COVID-19 episode analyzed in Section 5. The observed dynamics are consistent with the model’s interpretation of the episode, which predicts an increase in the consumption share of constrained households during the recession and a gradual normalization thereafter.²¹

²¹The measure can in principle also be constructed at monthly frequency, since CEX interviews take place every month and report expenditures separately for each reference month. Although potentially noisier, a monthly series could provide a more timely indicator of changes in the distribution of consumption during similar episodes.

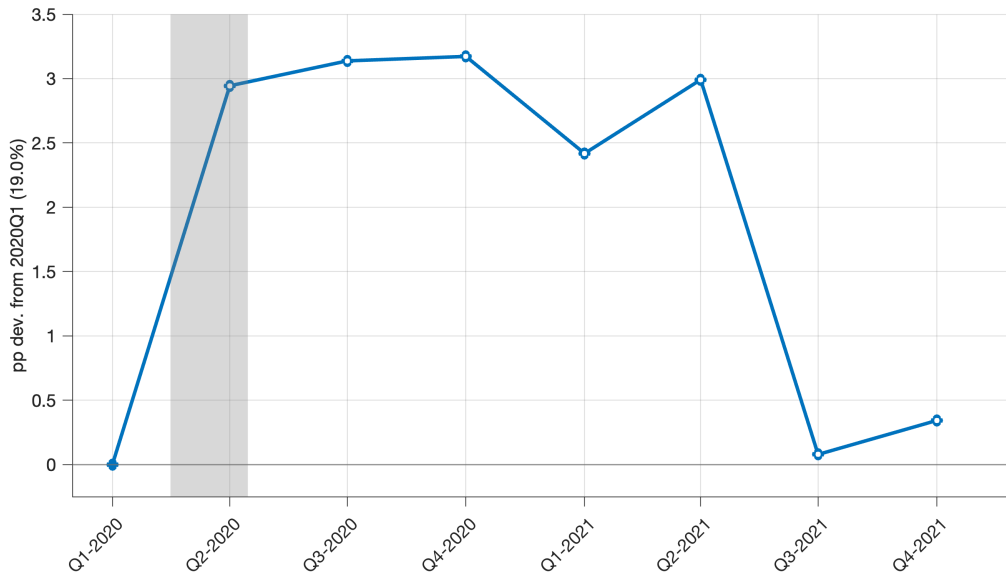


Figure 9: The hand-to-mouth consumption share through the pandemic

Note: The figure reports the share of aggregate nondurable and service expenditures accounted for by households below 200% of the federal poverty guideline, from 2020Q1 to 2021Q4. The series is expressed in percentage-point deviations from its 2020Q1 value.

Subcategory	UCC codes
Nondurables	
Apparel and footwear	360110, 360120, 360210, 360311, 360312, 360320, 360330, 360340, 360350, 360410, 360420, 360511, 360512, 360513, 360901, 360902, 370110, 370120, 370125, 370130, 370211, 370212, 370213, 370220, 370311, 370312, 370313, 370314, 370901, 370902, 370903, 370904, 380110, 380210, 380311, 380312, 380313, 380315, 380320, 380331, 380332, 380333, 380340, 380410, 380420, 380430, 380510, 380901, 380902, 380903, 390110, 390120, 390210, 390221, 390222, 390223, 390230, 390310, 390321, 390322, 390901, 390902, 400110, 400210, 400220, 400310, 410111, 410112, 410121, 410122, 410131, 410132, 410141, 410142, 410901, 410902, 410903, 410904, 410905, 420110, 420115, 420120
Household textiles, sewing materials	280110, 280120, 280130, 280140, 280220, 280230, 280900
Prescription, nonprescription drugs	540000, 550110
Reading materials	590110, 590111, 590112, 590210, 590211, 590212, 590310, 590410
Food, beverages on trips	790210, 790230, 790310, 790320, 790330
Other nondurable goods	320903, 330511, 610210, 610320, 630110, 630210, 640130, 640430, 720110, 720210, 990910
Energy: fuels	
Fuel oil, other household fuels	250111, 250112, 250113, 250114, 250211, 250212, 250213, 250214, 250221, 250222, 250223, 250224, 250901, 250902, 250903, 250904, 250911, 250912, 250913, 250914
Gasoline and motor oil	470111, 470112, 470113, 470211, 470212, 470220
Services	
Food, alcohol away from home	190901, 190902, 190903, 200900
Rent, lodging away from home	210110, 210210, 210310, 210901, 210902
Maintenance, repairs, property services	220611, 220615, 220900, 220901, 220902, 230111, 230112, 230113, 230114, 230115, 230116, 230119, 230121, 230122, 230123, 230131, 230134, 230141, 230142, 230150, 230151, 230152, 230901, 230902, 240111, 240121, 240211, 240221, 240311, 240321
Water, telephone, other public services	270000, 270101, 270102, 270103, 270104, 270105, 270106, 270211, 270212, 270213, 270214, 270310, 270311, 270411, 270412, 270413, 270414, 270510, 270610, 270901, 270902, 270903, 270904
Cable, satellite television	310240, 310243, 310350
Appliance repair and rental	320161, 320163, 320611, 320621, 320622, 320623, 320624, 320631, 320632, 320633
Household operations	340210, 340211, 340212, 340310, 340410, 340420, 340510, 340520, 340530, 340610, 340620, 340630, 340901, 340902, 340903, 340905, 340906, 340907, 340908, 340910, 340911, 340912, 340914, 340915

Table 7: Expenditure classification of the consumption aggregate

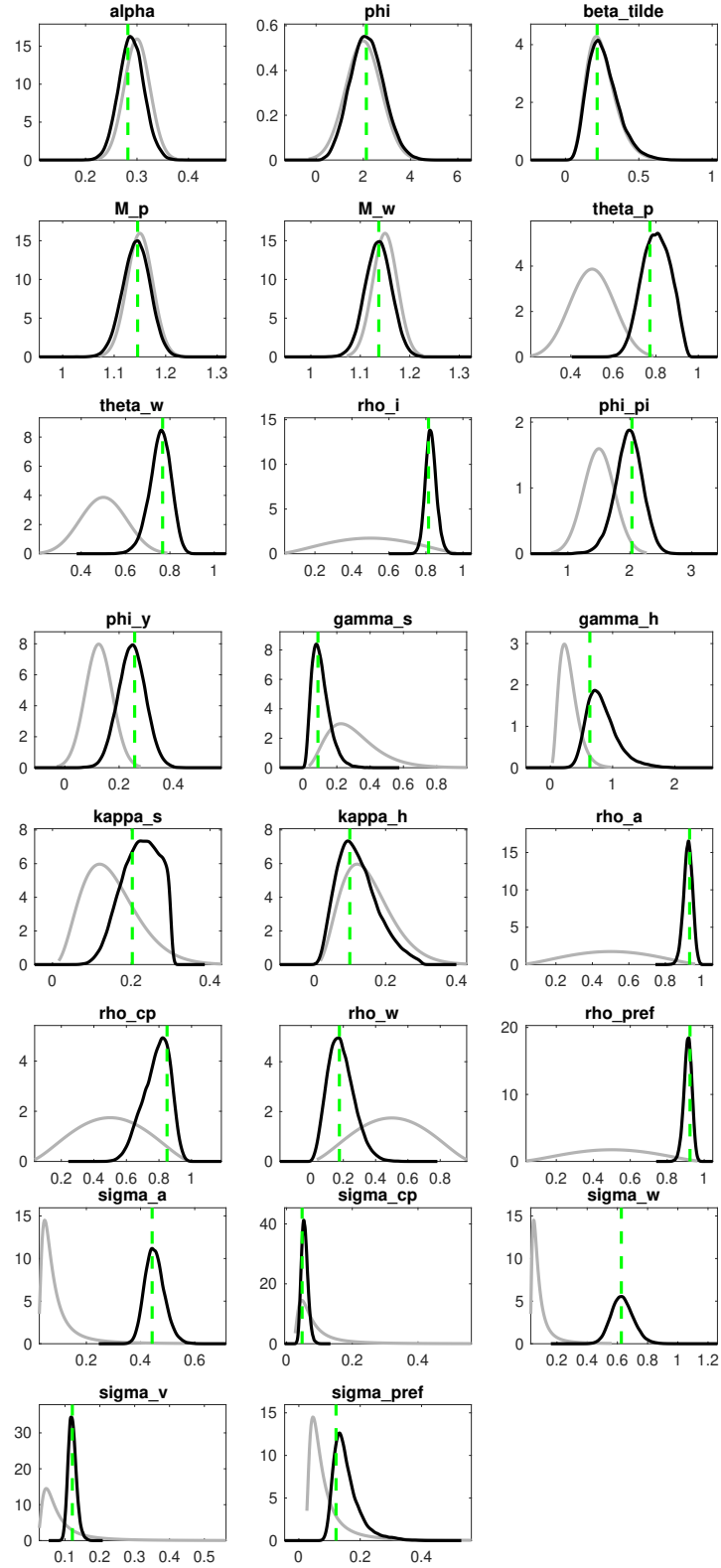
Subcategory	UCC codes
Services	
Rented vehicles	440110, 440120, 440130, 440140, 440150, 440210, 440900
Leased vehicles	450226, 450310, 450312, 450313, 450314, 450350, 450351, 450352, 450353, 450354, 450410, 450412, 450413, 450414, 450906
Vehicle maintenance and repair	460116, 460907, 460908, 460909, 490100, 490110, 490211, 490212, 490220, 490221, 490231, 490232, 490300, 490311, 490312, 490313, 490314, 490315, 490317, 490318, 490319, 490411, 490412, 490413, 490900
Registration, licenses, parking, tolls	520511, 520512, 520516, 520517, 520521, 520522, 520530, 520531, 520532, 520541, 520542, 520550, 520902, 520903, 520904, 520905, 520906, 520907
Public transportation	530110, 530210, 530311, 530312, 530411, 530412, 530510, 530901, 530902
Medical, dental, hospital services	560110, 560210, 560310, 560320, 560330, 560400, 560900, 570110, 570111, 570210, 570220, 570230, 570240, 570901, 570903
Entertainment fees and admissions	620111, 620115, 620122, 620211, 620212, 620213, 620214, 620221, 620222, 620310, 620320, 620330, 620410, 620420, 620902, 620903, 620904, 620905, 620906, 620907, 620908, 620909, 620912, 620916, 620917, 620918, 620919, 620921, 620922, 620926, 620930
Pet services, veterinary care	650110, 650210, 650310, 650900
Education	670110, 670210, 670310, 670410, 670901, 670902, 670903
Legal, financial, professional services	680110, 680140, 680210, 680310, 680901, 680902, 680903, 680904, 680905
Computer, information, recreation services	690113, 690114, 690116, 690310, 690320, 690330, 690340, 690350
Recreation, transportation on trips	790400, 790410, 790420, 790430, 790600, 790610, 790611, 790620, 790640, 790690, 790730, 790830, 790840, 790910
Other services	1400, 800700, 800710, 810301, 810302, 820301, 820302, 820401, 820402, 830101, 830102, 850216, 990900, 990920
Energy: utilities	
Electricity and natural gas	260111, 260112, 260113, 260114, 260211, 260212, 260213, 260214

Expenditure classification of the consumption aggregate (continued)

Note: The table reports the classification of Universal Classification Codes (UCCs) underlying the consumption aggregate in equation (D.1), grouped by expenditure subcategory. Consistent with the NIPA definitions, fuels are classified as nondurables and utilities as services. Durables are excluded because the model does not feature durable goods and durable purchases are typically treated as investment. Imputed-rent categories (UCCs beginning with 910), introduced in 2009, are also excluded to avoid a structural break in the service series.

D.2 Estimation results

Bayesian estimation



Estimation results: THANK. Table 8 reports the prior and posterior distributions of the THANK model, estimated on the same data and with the same priors as the D-THANK model, with the switching coefficients restricted to zero.

Parameter	Prior family	Prior mean	Prior s.d.	Post. mode	Post. mean	90% HPDI
<i>Preferences and technology</i>						
α	Normal	0.30	0.025	0.281	0.284	[0.243, 0.324]
φ	Normal	2.00	0.750	2.282	2.310	[1.202, 3.450]
$\tilde{\beta}$	Gamma	0.25	0.100	0.215	0.254	[0.091, 0.408]
$\mathcal{M}_p$	Normal	1.15	0.025	1.142	1.142	[1.098, 1.183]
$\mathcal{M}_w$	Normal	1.15	0.025	1.139	1.142	[1.095, 1.187]
θ_p	Beta	0.50	0.100	0.680	0.665	[0.580, 0.751]
θ_w	Beta	0.50	0.100	0.723	0.686	[0.601, 0.806]
<i>Monetary policy</i>						
ρ_i	Beta	0.50	0.200	0.798	0.794	[0.756, 0.834]
ϕ_π	Normal	1.50	0.250	2.031	2.073	[1.784, 2.370]
ϕ_y	Normal	0.125	0.050	0.264	0.262	[0.184, 0.340]
<i>Shock persistences</i>						
ρ_a	Beta	0.50	0.200	0.939	0.935	[0.895, 0.977]
ρ_{cp}	Beta	0.50	0.200	0.815	0.809	[0.738, 0.880]
ρ_w	Beta	0.50	0.200	0.176	0.233	[0.045, 0.364]
ρ_{pref}	Beta	0.50	0.200	0.941	0.935	[0.902, 0.968]
<i>Innovation standard deviations</i>						
σ_a	Inv. Gamma	0.10	2.000	0.442	0.451	[0.394, 0.508]
σ_{cp}	Inv. Gamma	0.10	2.000	0.066	0.075	[0.052, 0.094]
σ_w	Inv. Gamma	0.10	2.000	0.667	0.663	[0.524, 0.808]
σ_v	Inv. Gamma	0.10	2.000	0.126	0.130	[0.109, 0.150]
σ_{pref}	Inv. Gamma	0.10	2.000	0.095	0.102	[0.076, 0.126]

Table 8: Prior and posterior distributions of the THANK model

Note: The THANK model imposes $\gamma_s = \gamma_h = \kappa_s = \kappa_h = 0$, so the household composition is fixed. Priors on the remaining parameters are identical to those of the D-THANK model in Table 1. Posterior mode, mean, and 90% highest posterior density interval (HPDI) are computed from 500,000 draws of the random-walk Metropolis–Hastings sampler with 20% burn-in. Sample: 1987Q1–2007Q4.

Historical share of hand-to-mouth households. Several studies have examined hand-to-mouth behavior using cross-sectional survey data on household portfolios (e.g., Aguiar et al., 2024; Kaplan et al., 2014). Nevertheless, constructing a reliable time se-

ries of the hand-to-mouth household share remains challenging for several reasons. First, measurements of hand-to-mouth behavior are highly sensitive to how households' net worth is classified (e.g., wealthy vs. non-wealthy hand-to-mouth). Additionally, because income data are typically reported on an annual basis, higher-frequency time series are likely to reflect changes only gradually.²²

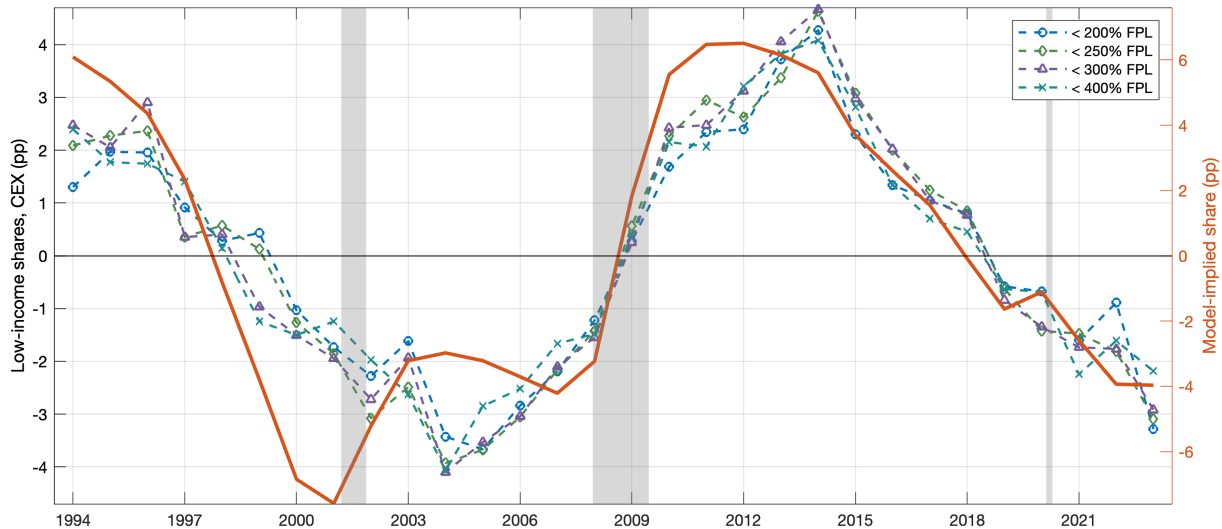


Figure 10: Model-implied vs. observed historical share of hand-to-mouth households

Note: The dashed lines show the quadratically detrended share of households whose before-tax family income over the preceding twelve months falls below 200%, 250%, 300%, and 400% of the federal poverty guideline (HHS/ASPE), as measured in the Consumer Expenditure Survey, each with a distinct marker. The blue solid line shows the smoothed model-implied hand-to-mouth share, expressed in annual terms and quadratically detrended for comparison with the observed shares.

In contrast, a model-based approach provides a compelling alternative by generating internally consistent estimates of variables for which no consensus empirical measure exists (e.g., the hand-to-mouth share). These estimates are obtained by confronting the estimated model, with all parameters fixed at their posterior modes from the 1987Q1–2007Q4 sample, with data rebuilt over an extended window; the portion of the model-implied series after 2007 therefore lies strictly outside the estimation sample. Figure 10 presents an example of this approach, comparing the model-implied historical share of hand-to-mouth households with empirical proxies based on reported before-tax income.²³ As illustrated in the figure, the

²²This is due to the retrospective nature of surveys, where households report income over the preceding twelve months during each quarterly interview.

²³While income characteristics are not the sole determinant of hand-to-mouth behavior, they serve as an important proxy for consumption patterns, liquidity constraints, and financial vulnerability.

model-implied share of hand-to-mouth households closely aligns with the empirical proxies, displaying similar cyclical dynamics – especially in response to economic downturns such as the early 2000s recession and the Great Recession. Interestingly, the model-implied series responds more rapidly to income fluctuations, whereas survey-based measures reflect these changes with a lag due to retrospective reporting. Overall, these findings are consistent with the income elasticity estimates derived from Equation (2.16), reinforcing their accord with observed cyclical patterns.

E Welfare function

This appendix derives a second-order approximation to the Ramsey planner's objective, adapting the approach of McKay and Wolf (2023) to a two-agent economy in which the composition of households is itself time-varying.

Because the mass of each household type changes over time, it is convenient to work with the total consumption of each group rather than with per-capita consumption. Define

$$\tilde{C}_t^H \equiv \lambda_t C_t^H, \quad \tilde{C}_t^S \equiv (1 - \lambda_t) C_t^S \quad (\text{E.1})$$

so that $\tilde{C}_t^i$ is the consumption accruing to group i as a whole. By the aggregation identity, $C_t^A = \tilde{C}_t^H + \tilde{C}_t^S$, and the consumption share of group i is thus

$$\omega_{it} \equiv \frac{\tilde{C}_t^i}{C_t^A}, \quad \text{so that} \quad \sum_{i=\{H,S\}} \omega_{it} = 1 \quad \text{for all } t. \quad (\text{E.2})$$

The planner's objective. The planner values the consumption of each group, weighted by a Pareto weight φ_i .²⁴

$$U_t = \sum_{i=\{H,S\}} \varphi_i \left(\frac{(\tilde{C}_{it})^{1-\sigma}}{1-\sigma} - \frac{(N_t)^{1+\varphi}}{1+\varphi} \right). \quad (\text{E.3})$$

Using $\tilde{C}_{it} = \omega_{it} C_t^A$ and expressing aggregate consumption and hours in deviations from steady state, $C_t^A = e^{\hat{c}_t^A} C^A$ and $N_t = e^{\hat{n}_t} N$, the objective becomes

$$U_t = \sum_{i=\{H,S\}} \varphi_i \left(\frac{(\omega_{it} e^{\hat{c}_t^A} C^A)^{1-\sigma}}{1-\sigma} - \frac{(e^{\hat{n}_t} N)^{1+\varphi}}{1+\varphi} \right). \quad (\text{E.4})$$

The goal is to approximate (E.4) to second order.

²⁴Because both types supply the same hours, aggregating over the two groups leaves the labor term unchanged, since $\lambda_t N_t + (1 - \lambda_t) N_t = N_t$.

Steady state and Pareto weights. Efficiency in the steady state requires that weighted marginal utilities of consumption be equalized across groups:

$$\varphi_i \bar{\omega}_i^{-\sigma} (C^A)^{1-\sigma} = \bar{u}_c C^A \quad \forall i \quad (\text{E.5})$$

where $\bar{u}_c$ denotes the common marginal value of consumption. Rearranging (E.5) and normalizing by the sum across groups gives the steady-state consumption shares:

$$\bar{\omega}_i = \frac{\varphi_i^{1/\sigma}}{\sum_{j \in \{H,S\}} \varphi_j^{1/\sigma}} \quad \forall i. \quad (\text{E.6})$$

Here, the Pareto weights are assigned residually to support the efficient steady-state allocation of consumption across groups. For future reference, define also

$$\Xi \equiv \left(\sum_{i \in \{H,S\}} \varphi_i^{1/\sigma} \right)^\sigma = \bar{u}_c (C^A)^\sigma = \varphi_i \bar{\omega}_i^{-\sigma} \quad \forall i. \quad (\text{E.7})$$

Second-order approximation. With the steady-state shares and the constant Ξ in hand, expanding (E.4) around the steady state yields

$$\begin{aligned} U_t \approx & \bar{U} + (C^A)^{1-\sigma} \Xi \hat{c}_t^A - N^{1+\varphi} \hat{n}_t + \frac{1}{2} (1-\sigma) (C^A)^{1-\sigma} \Xi (\hat{c}_t^A)^2 \\ & - \frac{1}{2} (1+\varphi) N^{1+\varphi} \hat{n}_t^2 - \frac{1}{2} \sigma (C^A)^{1-\sigma} \Xi \sum_{i \in \{H,S\}} \frac{\hat{\omega}_{it}^2}{\bar{\omega}_i} \\ & + (C^A)^{1-\sigma} \Xi \sum_{i \in \{H,S\}} \hat{\omega}_{it} + (1-\sigma) (C^A)^{1-\sigma} \Xi \hat{c}_t^A \sum_{i \in \{H,S\}} \hat{\omega}_{it}. \end{aligned} \quad (\text{E.8})$$

However, the last two terms are zero because $\sum_i \hat{\omega}_{it} = 0$, so that

$$\begin{aligned} U_t \approx & \bar{U} + (C^A)^{1-\sigma} \Xi \hat{c}_t^A - N^{1+\varphi} \hat{n}_t + \frac{1}{2} (1-\sigma) (C^A)^{1-\sigma} \Xi (\hat{c}_t^A)^2 \\ & - \frac{1}{2} (1+\varphi) N^{1+\varphi} \hat{n}_t^2 - \frac{1}{2} \sigma (C^A)^{1-\sigma} \Xi \sum_{i \in \{H,S\}} \frac{\hat{\omega}_{it}^2}{\bar{\omega}_i}. \end{aligned} \quad (\text{E.9})$$

It remains to express (E.9) in terms of output, inflation, and consumption shares.

Rewriting the resource constraint. First, we need to establish a relationship between aggregate employment and output. Recall that aggregate employment is obtained by integrating labor across skills z and products j :

$$N_t = \int_0^1 \int_0^1 N_t(j, z) dz dj = \Delta_{w,t} \Delta_{p,t} \left(\frac{Y_t}{A_t} \right)^{\frac{1}{1-\alpha}} \quad (\text{E.10})$$

where $\Delta_{p,t} \equiv \int_0^1 (Y_t(j) / Y_t)^{\frac{1}{1-\alpha}} dj$ and $\Delta_{w,t} \equiv \int_0^1 (N_t(j, z) / N_t(j)) dz$ measure price and wage dispersion. Taking logs,

$$(1 - \alpha) \hat{n}_t = \hat{y}_t - a_t + d_{w,t} + d_{p,t} \quad (\text{E.11})$$

with $d_{p,t} \equiv (1 - \alpha) \log \int_0^1 (P_t(j) / P_t)^{-\frac{\varepsilon_p}{1-\alpha}} dj$ and $d_{w,t} \equiv (1 - \alpha) \log \int_0^1 (W_t(z) / W_t)^{-\varepsilon_w} dz$.

Following Gali (2015), second-order expansions of the price and wage indices give

$$d_{p,t} \approx \frac{\varepsilon_p}{2\Theta} \text{var}_j \{p_t(j)\}, \quad d_{w,t} \approx \frac{(1 - \alpha) \varepsilon_w}{2} \text{var}_z \{w_t(z)\} \quad (\text{E.12})$$

where $\Theta \equiv \frac{1-\alpha}{1-\alpha+\alpha\varepsilon_p}$. Finally, combining these approximations with (E.11) yields

$$\hat{n}_t = \frac{\hat{y}_t - a_t}{1 - \alpha} + \frac{\varepsilon_w}{2} \text{var}_z \{w_t(z)\} + \frac{\varepsilon_p}{2\Theta(1 - \alpha)} \text{var}_j \{p_t(j)\}. \quad (\text{E.13})$$

The welfare loss function. With hours expressed in terms of output and dispersion, substitute (E.13) into (E.9) and use the goods-market clearing condition $\hat{c}_t^A = \hat{y}_t$ to get

$$\begin{aligned}
U_t \approx \bar{U} + (C^A)^{1-\sigma} \Xi \hat{y}_t \\
- N^{1+\varphi} \left(\frac{\hat{y}_t - a_t}{1-\alpha} + \frac{\varepsilon_w}{2} \text{var}_z \{w_t(z)\} + \frac{\varepsilon_p}{2\Theta(1-\alpha)} \text{var}_j \{p_t(j)\} \right) \\
+ \frac{1}{2} (1-\sigma) (C^A)^{1-\sigma} \Xi \hat{y}_t^2 - \frac{1}{2} \sigma (C^A)^{1-\sigma} \Xi \sum_{i=\{H,S\}} \frac{\hat{\omega}_{it}^2}{\bar{\omega}_i} \\
- \frac{1}{2} (1+\varphi) N^{1+\varphi} \left(\frac{\hat{y}_t - a_t}{1-\alpha} + \frac{\varepsilon_w}{2} \text{var}_z \{w_t(z)\} + \frac{\varepsilon_p}{2\Theta(1-\alpha)} \text{var}_j \{p_t(j)\} \right)^2. \quad (\text{E.14})
\end{aligned}$$

Discarding terms of order higher than second, together with terms independent of policy, reduces (E.14) to

$$\begin{aligned}
U_t \approx \bar{U} + (C^A)^{1-\sigma} \Xi \hat{y}_t - \frac{N^{1+\varphi}}{1-\alpha} \hat{y}_t - N^{1+\varphi} \frac{\varepsilon_w}{2} \text{var}_z \{w_t(z)\} \\
- N^{1+\varphi} \frac{\varepsilon_p}{2\Theta(1-\alpha)} \text{var}_j \{p_t(j)\} + \frac{1}{2} (1-\sigma) (C^A)^{1-\sigma} \Xi \hat{y}_t^2 \\
- \frac{1}{2} \frac{(1+\varphi) N^{1+\varphi}}{(1-\alpha)^2} \hat{y}_t^2 - \frac{1}{2} \sigma (C^A)^{1-\sigma} \Xi \sum_{i=\{H,S\}} \frac{\hat{\omega}_{it}^2}{\bar{\omega}_i}. \quad (\text{E.15})
\end{aligned}$$

Returning to the planner's problem, under the assumption that $\sigma = 1$, the discounted sum of utility then reads

$$\begin{aligned}
\sum_{t=0}^{\infty} \beta^t (U_t - \bar{U}) \approx -\frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left[N^{1+\varphi} \varepsilon_w \text{var}_z \{w_t(z)\} + N^{1+\varphi} \frac{\varepsilon_p}{\Theta(1-\alpha)} \text{var}_j \{p_t(j)\} \right. \\
\left. + \frac{(1+\varphi) N^{1+\varphi}}{(1-\alpha)^2} \hat{y}_t^2 + \Xi \sum_{i=\{H,S\}} \frac{\hat{\omega}_{it}^2}{\bar{\omega}_i} \right] \quad (\text{E.16})
\end{aligned}$$

where the linear term is zero, since in the efficient steady state $\bar{u}_c (1-\alpha) N^{-\alpha} = N^\varphi$. Following Gali (2015), the discounted sums of the price and wage dispersion terms map into squared inflation through the Calvo pricing structure, and the efficiency condition implies

$\Xi = \bar{u}_c C^A = \frac{N^{1+\varphi}}{1-\alpha}$, so that (E.16) becomes

$$\sum_{t=0}^{\infty} \beta^t (U_t - \bar{U}) \approx -\frac{1}{2} N^{1+\varphi} \sum_{t=0}^{\infty} \beta^t \left[\kappa_w (\pi_t^w)^2 + \frac{\kappa_p}{\Theta(1-\alpha)} (\pi_t^p)^2 + \frac{1+\varphi}{(1-\alpha)^2} \hat{y}_t^2 + \frac{1}{1-\alpha} \sum_{i=\{H,S\}} \frac{\hat{\omega}_{it}^2}{\bar{\omega}_i} \right]$$

where $\kappa_w \equiv \frac{\theta_w \varepsilon_w}{(1-\beta\theta_w)(1-\theta_w)}$ and $\kappa_p \equiv \frac{\theta_p \varepsilon_p}{(1-\beta\theta_p)(1-\theta_p)}$. Finally, normalizing by $\bar{u}_c C^A$ delivers

$$\sum_{t=0}^{\infty} \beta^t \left(\frac{U_t - \bar{U}}{\bar{u}_c C^A} \right) \approx -\mathcal{L} = -\frac{1}{2} \sum_{t=0}^{\infty} \beta^t \mathcal{L}_t \quad (\text{E.17})$$

where the per-period loss is

$$\mathcal{L}_t = \underbrace{(1-\alpha) \kappa_w (\pi_t^w)^2}_{\chi_w} + \underbrace{\frac{1-\alpha+\alpha\varepsilon_p}{1-\alpha} \kappa_p (\pi_t^p)^2}_{\chi_p} + \underbrace{\frac{1+\varphi}{1-\alpha} \hat{y}_t^2}_{\chi_y} + \underbrace{\frac{1}{\bar{\omega}_S} \hat{\omega}_{S,t}^2}_{\chi_S} + \underbrace{\frac{1}{\bar{\omega}_H} \hat{\omega}_{H,t}^2}_{\chi_H}. \quad (\text{E.18})$$

Consumption shares. The loss function penalizes deviations of the consumption shares from their steady-state values. Log-linearizing (E.2), these deviations are

$$\hat{\omega}_{H,t} = \bar{\omega}_H \left(\frac{1}{\lambda} \hat{\lambda}_t + \hat{c}_t^H - \hat{c}_t^A \right), \quad (\text{E.19})$$

$$\hat{\omega}_{S,t} = \bar{\omega}_S \left(-\frac{1}{1-\lambda} \hat{\lambda}_t + \hat{c}_t^S - \hat{c}_t^A \right). \quad (\text{E.20})$$

Note that in the D-THANK, each expression now contains two distinct terms. The first captures the change in the size of each group of households through $\hat{\lambda}_t$, and is absent in models with a fixed household distribution (e.g., TANK, THANK). The second captures the consumption of each household type *relative* to the aggregate. As discussed in the main text, this implies that the planner's objective extends beyond stabilizing relative consumption to also stabilizing the composition of households.

Consumption equivalent variation. Expressing welfare in utils can be difficult to interpret. A more transparent metric is the consumption equivalent variation, which converts welfare differences into consumption units.

Let $\mathcal{W}^p$ and $\mathcal{W}^b$ denote welfare under a given policy p and under a benchmark b . From the derivations above, $\mathcal{W}^p \propto -\mathcal{L}^p$. Consider now augmenting consumption under the benchmark by a constant fraction δ at every date:

$$\mathcal{W}^b(\delta) = \sum_{t=0}^{\infty} \beta^t \sum_{i=\{H,S\}} \varphi_i \left(\log \left(\omega_{it}^b C_t^{A,b} (1 + \delta) \right) - \frac{(N_t^b)^{1+\varphi}}{1 + \varphi} \right). \quad (\text{E.21})$$

Because $\sigma = 1$, the scaling factor separates additively:

$$\mathcal{W}^b(\delta) = \frac{\log(1 + \delta)}{1 - \beta} + \mathcal{W}^b(0) \propto \frac{\log(1 + \delta)}{1 - \beta} - \mathcal{L}^b. \quad (\text{E.22})$$

The consumption equivalent variation is the value of δ that equates welfare under the two regimes, $\mathcal{W}^p = \mathcal{W}^b(\delta)$. Solving,

$$\delta^* = \exp \left[(1 - \beta) (\mathcal{L}^b - \mathcal{L}^p) \right] - 1 \quad (\text{E.23})$$

which expresses the welfare gain of policy p over the benchmark as a permanent percentage change in consumption.

F Inequality in the D-THANK

The welfare analysis of Section 4 shows that the consumption gap between hand-to-mouth households and savers is one of the two sources of distributional fluctuations in the planner’s objective. This appendix characterizes the determinants of this gap and its cyclical behavior. Following Bilbiie (2024), the analysis is conducted in terms of the income gap, which is the primitive object in the present framework and maps directly into consumption inequality. Throughout, equation numbers refer to the log-linear system reported in Table 4. For expositional simplicity, I impose flexible wages ($\theta_w \rightarrow 0$) and set $\alpha = 0$.²⁵

A simple characterization of inequality. Define income inequality as the difference between saver and hand-to-mouth income,

$$\gamma_t \equiv \hat{y}_t^S - \hat{y}_t^H. \quad (\text{F.1})$$

In the zero-liquidity equilibrium, households consume their current income, implying $\hat{c}_t^i = \hat{y}_t^i$. Consequently, γ_t also measures consumption inequality and coincides with the negative of the consumption gap of the main text, $\gamma_t = -\hat{c}_t^G$. Characterizing the income gap therefore requires expressing each household type’s income as a function of aggregate variables.

Hand-to-mouth households earn only labor income and hold no financial assets, so that

$$\hat{y}_t^H = \hat{\omega}_t + \hat{y}_t, \quad (\text{F.2})$$

where the production function (eq. 9) has been used to eliminate hours, $\hat{n}_t$. Saver income, in turn, follows from the aggregate income identity (eq. 10),

$$\hat{y}_t^S = \psi_S^{-1} \hat{y}_t - \psi_S^{-1} \psi_H \hat{y}_t^H + \psi_S^{-1} \psi_\lambda \hat{\lambda}_t. \quad (\text{F.3})$$

²⁵Setting $\alpha = 0$ is without loss of generality for the qualitative results. The income gap remains countercyclical for every $\alpha \in [0, 1)$, because its cyclical coefficient is proportional to $\lambda \mathcal{M}_p + (1 - \alpha)(1 - 2\lambda) > 0$ throughout the admissible parameter space.

Combining these expressions and using $\psi_H + \psi_S = 1$ gives

$$\gamma_t = -\psi_S^{-1}\hat{\omega}_t + \psi_S^{-1}\psi_\lambda\hat{\lambda}_t \quad (\text{F.4})$$

so that the income gap is driven by two forces, fluctuations in the real wage and changes in household composition.

It remains to express the real wage in terms of aggregates. Substituting the wage-setting equation (eq. 1) and the wage Phillips curve (eq. 2) into (F.4), eliminating household incomes, and rearranging yields

$$\gamma_t = \Pi_y\hat{y}_t + \Pi_\lambda\hat{\lambda}_t, \quad (\text{F.5})$$

where

$$\Pi_y = -\frac{1+\varphi}{1-\lambda}[\lambda\Gamma + (1-\lambda)] < 0, \quad \Pi_\lambda = \frac{\Gamma-1}{1-\lambda} > 0,$$

and $\Gamma \equiv \frac{Y^S}{Y^H} = \frac{\mathcal{M}_p - \lambda}{1-\lambda}$ denotes the steady-state income ratio between savers and hand-to-mouth households. Equation (F.5) decomposes inequality into an *aggregate* and a *composition* channel. As in the THANK, unequal profit ownership generates income dispersion through the steady-state income differential Γ . Unlike the THANK, however, endogenous fluctuations in household composition provide an additional source of inequality through changes in $\hat{\lambda}_t$.

The compositional amplification of inequality. The decomposition in (F.5) also explains the cyclical behavior of inequality. Holding household composition fixed, the aggregate channel ($\Pi_y < 0$) reproduces the mechanism emphasized by Bilbiie (2024). Because hand-to-mouth households rely exclusively on labor income, whereas saver income combines labor and profit income, hand-to-mouth income is more cyclical. As a result, the income gap widens in recessions and narrows in expansions. The D-THANK introduces a second mechanism through the composition channel ($\Pi_\lambda > 0$). As households transition between saver and hand-to-mouth status over the business cycle, changes in household composition become

an additional source of cyclical variation in inequality. Differentiating (F.5) with respect to output gives

$$\frac{d\gamma_t}{d\hat{y}_t} = \Pi_y + \Pi_\lambda \Lambda_y, \quad (\text{F.6})$$

where $\Lambda_y \equiv \frac{\partial \hat{\lambda}_t}{\partial \hat{y}_t} < 0$ follows from the switching rule (eqs. 13–15), since the share of hand-to-mouth households declines during expansions. Because $\Pi_\lambda > 0$ and $\Lambda_y < 0$, endogenous household transitions *amplify* the cyclical variation in inequality. Relative to an otherwise identical economy with fixed household composition, aggregate disturbances affect the income gap not only through changes in factor incomes but also through endogenous changes in household composition. In terms of the main-text object, the countercyclical variation in γ_t implies that the consumption gap $\hat{c}_t^G$ declines in downturns, consistent with the countercyclical consumption share documented in Section 5.

G D-THANK with liquidity

This appendix develops a version of the D-THANK model with positive liquidity. Following Bilbiie (2024), the zero-liquidity assumption introduced in Section 2 is relaxed, allowing savers to hold a non-zero quantity of government bonds in equilibrium. Relative to the baseline model, only two blocks are affected. First, the budget constraint of hand-to-mouth households now includes a bond windfall from savers transitioning into the constrained state. Second, the government budget constraint is extended to account for the joint determination of transfers, taxes, and public debt, with fiscal policy governed by a debt-consolidation rule in the spirit of Gali (2020). This extension allows the government to finance transfers through debt issuance rather than contemporaneous taxation. The remaining equilibrium conditions are identical to those in the baseline model and are not repeated here.

G.1 Households

Relaxing the zero-liquidity assumption allows savers to hold government bonds in equilibrium. Since hand-to-mouth households continue to satisfy $Z_{t+1}^H = 0$, bond market clearing implies that savers hold the entire stock of one-period government bonds,

$$(1 - \lambda_t)Z_{t+1}^S = B_{t+1}.$$

The wealth-accumulation equation (2.5) then implies that the financial wealth carried into the constrained state is

$$\mathcal{B}_{t+1}^H = (1 - \lambda_t)(1 - s_{t+1})Z_{t+1}^S = (1 - s_{t+1})B_{t+1}, \quad (\text{G.1})$$

which is no longer zero because savers transitioning into the constrained state retain their bond holdings. Consequently, the only modification to the household block concerns the

budget constraint of hand-to-mouth households, which becomes

$$C_t^H = \frac{W_t}{P_t} N_t + T_t^H + \frac{1 + i_{t-1}}{1 + \pi_t} \frac{(1 - s_t)}{\lambda_t} B_t \equiv Y_t^D, \quad (\text{G.2})$$

where the final term captures the bond windfall received by former savers, making outstanding public debt a source of disposable income for households transitioning into the constrained state.

G.2 Government

With bonds in positive supply, the government must service its outstanding debt in addition to financing transfers, and the budget need no longer balance in every period. The balanced-budget condition (A.12) of the baseline model is therefore replaced by the consolidated government budget constraint,

$$B_{t+1} = \frac{1 + i_{t-1}}{1 + \pi_t} B_t + \lambda_t T_t^H + (1 - \lambda_t) T_t^S, \quad (\text{G.3})$$

under which transfers, taxes, and debt issuance are jointly determined. To ensure that debt returns to its long-run level, aggregate taxes on savers follow a consolidation rule in the spirit of Gali (2020),

$$(1 - \lambda_t) T_t^S = -\phi_b B_t + \tilde{T}_t^S, \quad (\text{G.4})$$

where $\phi_b > 0$ governs the speed of consolidation and $\tilde{T}_t^S$ is a discretionary component that specifies the financing regime. Under tax financing, $\tilde{T}_t^S = -\lambda_t T_t^H$: taxes on savers offset the transfer contemporaneously, the budget balances in every period, and the baseline model is recovered. Under debt financing, $\tilde{T}_t^S = 0$: the transfer is financed through debt issuance, and the outstanding stock is retired gradually through the consolidation component levied on savers.

G.3 Log-linear system

Linearizing (G.2)–(G.4) around the steady state,²⁶ only the disposable-income equation is modified relative to the log-linearized system reported in Table 4:

$$\hat{y}_t^D = \hat{\omega}_t + \hat{n}_t + \frac{\mathcal{M}_p}{1 - \alpha} \hat{t}_t^H + \frac{\mathcal{M}_p}{1 - \alpha} \frac{(1 + i^*)(1 - s)}{\lambda} \hat{b}_t, \quad (\text{G.5})$$

where $\hat{t}_t^H$, $\hat{t}_t^S$, and $\hat{b}_t$ denote transfers, taxes, and public debt, respectively, expressed as ratios to steady-state output. The system is then closed by the law of motion for public debt, obtained by combining the government budget constraint (G.3) with the fiscal rule (G.4),

$$\hat{b}_{t+1} = (1 + i^* - \phi_b) \hat{b}_t + \lambda \hat{t}_t^H + \hat{t}_t^S, \quad (\text{G.6})$$

which is stable provided $\phi_b > i^*$. All remaining equations in Table 4 are unchanged. The financing regimes are defined analogously, with tax financing setting $\hat{t}_t^S = -\lambda \hat{t}_t^H$, under which the baseline model is recovered, and debt financing setting $\hat{t}_t^S = 0$. Throughout the quantitative analysis, ϕ_b is set to 0.02, consistent with the slow-consolidation calibration of Gali (2020).

²⁶With little loss of generality, the steady state is assumed to feature zero public debt. Allowing for a positive steady-state debt stock increases the numerical sensitivity of the model solution, while the additional first-order terms it introduces play a limited role in the model's quantitative implications. Specifically, the terms associated with household transitions and household composition largely offset one another, and real interest rate fluctuations are small at business-cycle frequencies.

H Impulse response functions

H.1 Responses to Other Exogenous Shocks

This appendix reports the responses to the productivity, monetary policy, wage-push, and preference shocks, each comparing the D-THANK model (solid blue) and the THANK model (dashed orange).

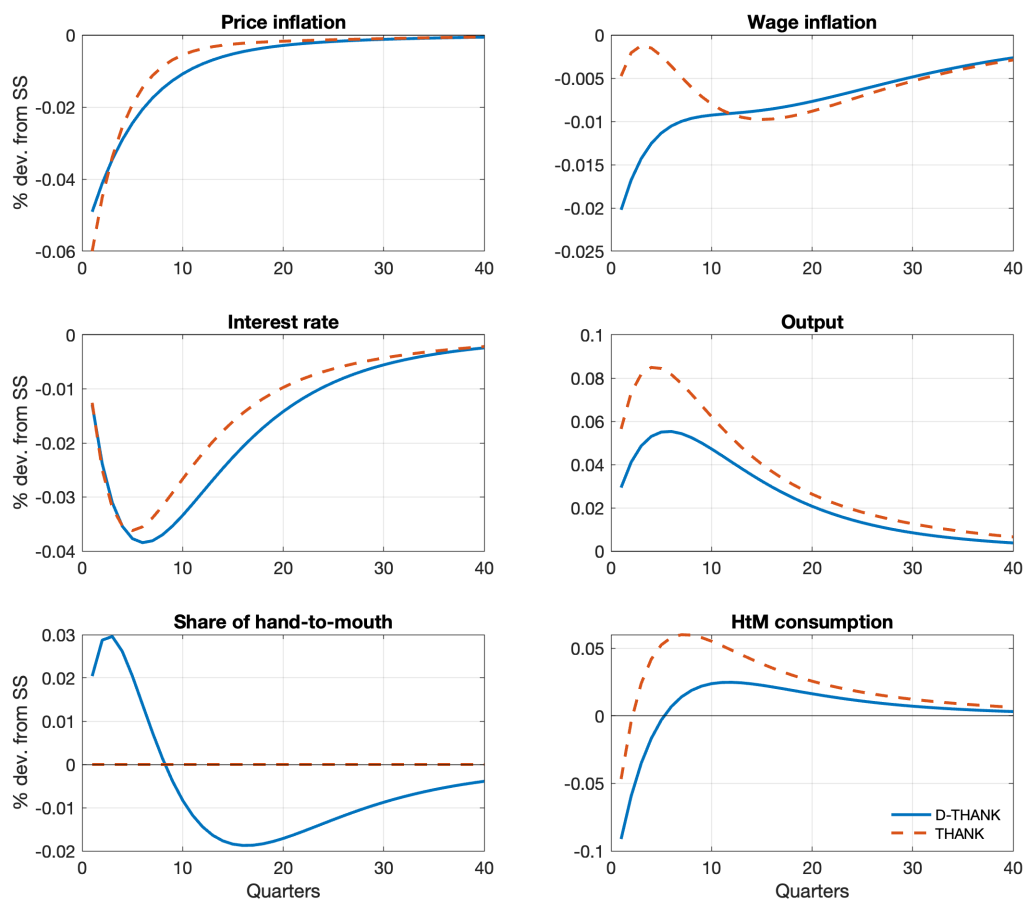


Figure 11: Impulse response functions to a productivity shock

Note: Impulse responses are constructed at the posterior mode of the structural parameters. Both models are normalized to a 0.1 percent innovation to total factor productivity on impact. Solid blue lines refer to the D-THANK model and dashed orange lines to the THANK model.

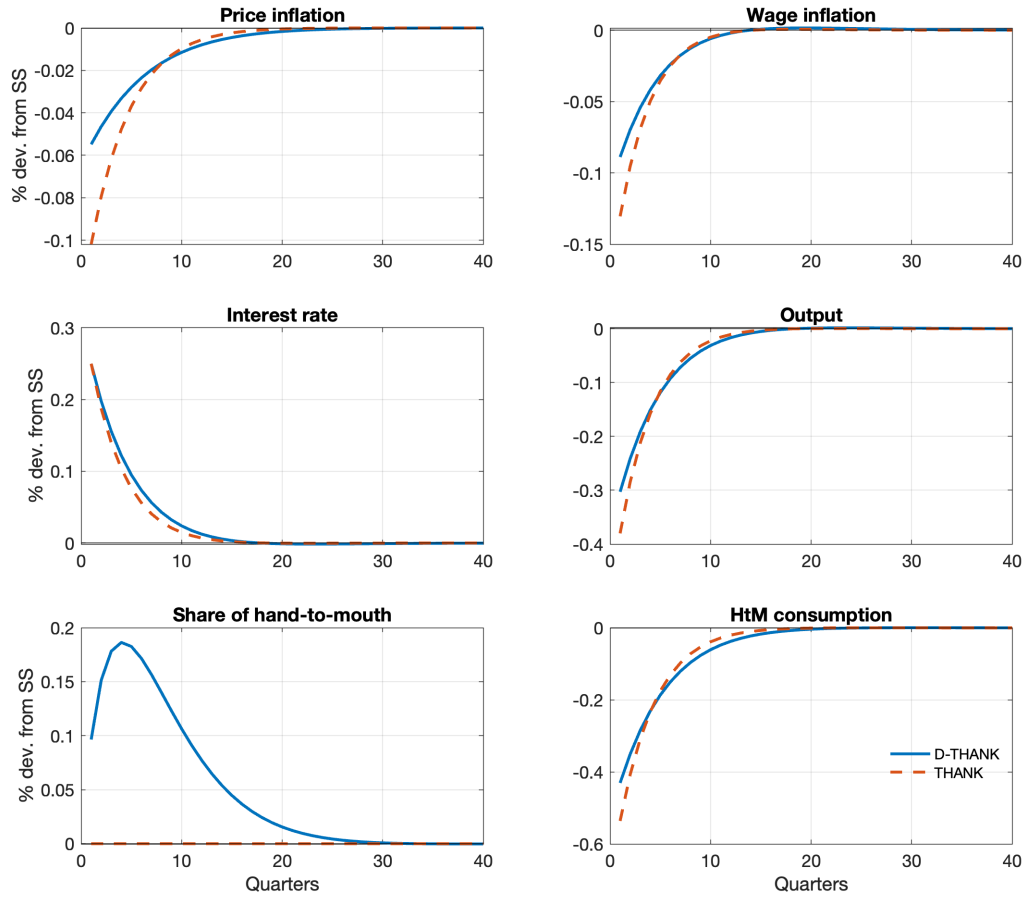


Figure 12: Impulse response functions to a monetary policy shock

Note: Impulse responses are constructed at the posterior mode of the structural parameters. Both models are normalized to a 0.25 percent annualized innovation to the nominal interest rate on impact. Solid blue lines refer to the D-THANK model and dashed orange lines to the THANK model.

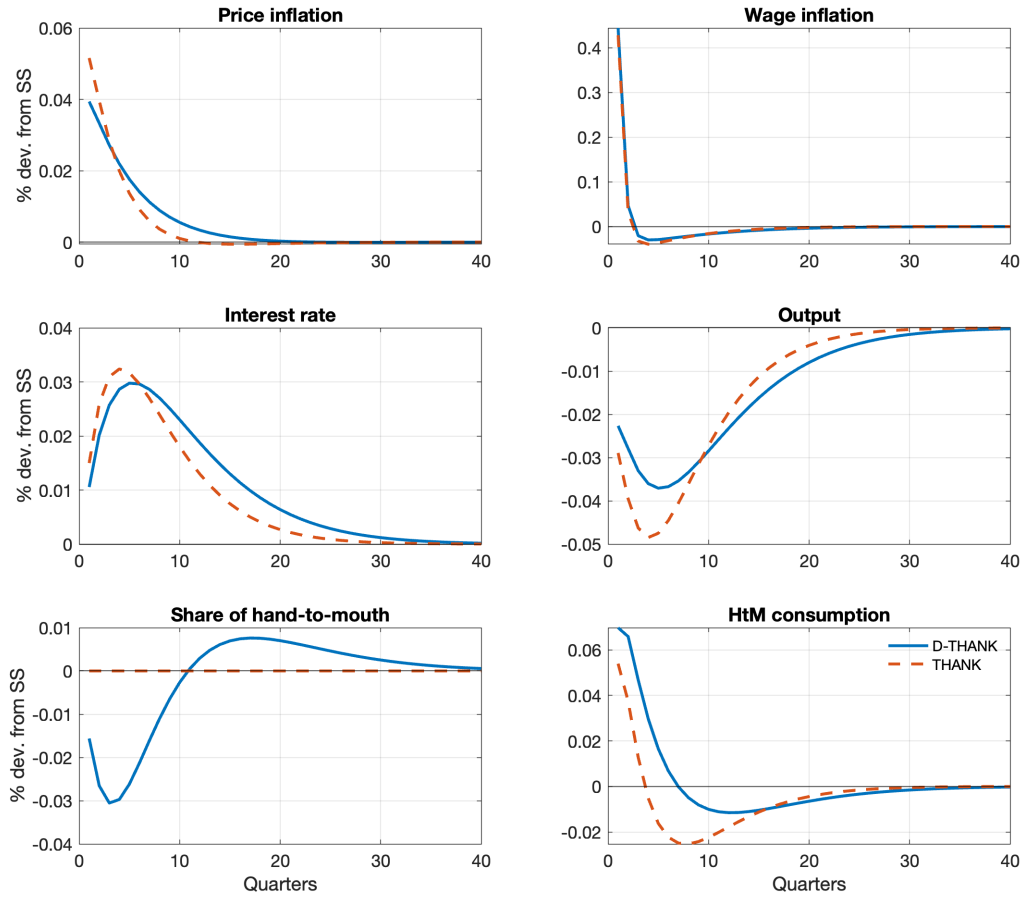


Figure 13: Impulse response functions to a wage-push shock

Note: Impulse responses are constructed at the posterior mode of the structural parameters. Both models are normalized to a 0.1 percent innovation to the wage-push process on impact. Solid blue lines refer to the D-THANK model and dashed orange lines to the THANK model.

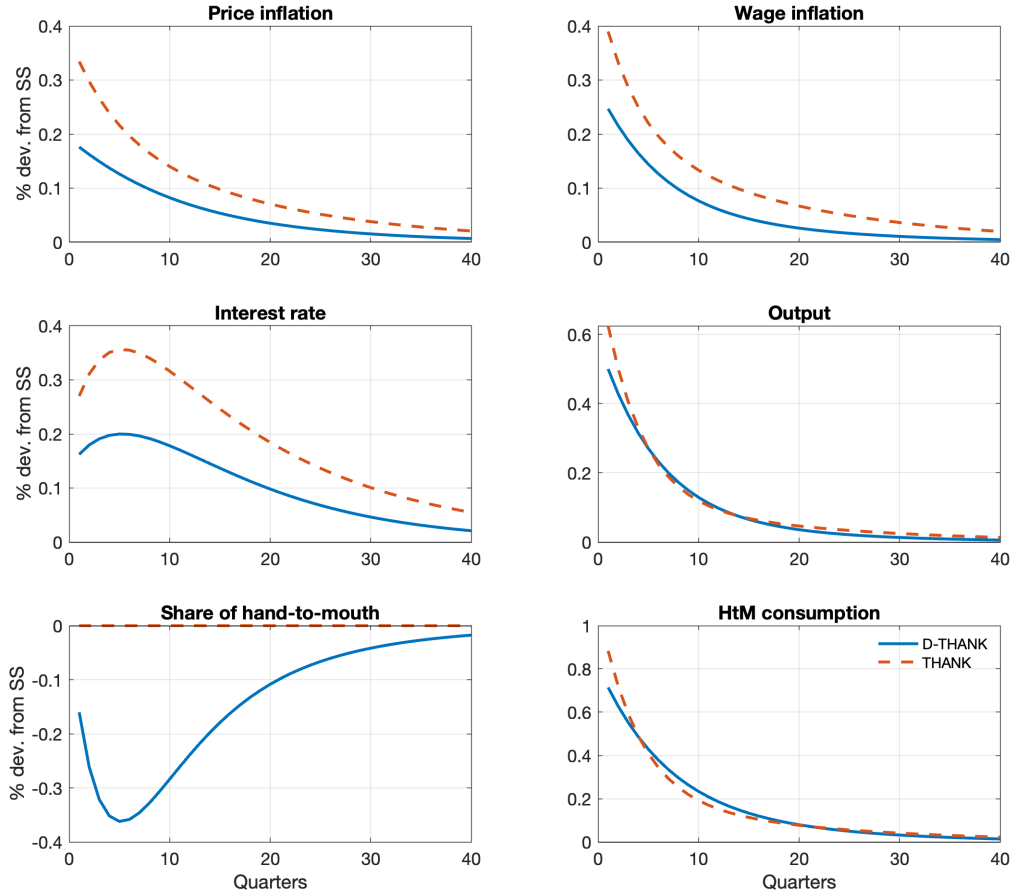


Figure 14: Impulse response functions to a preference shock

Note: Impulse responses are constructed at the posterior mode of the structural parameters. Both models are normalized to a 0.1 percent innovation to the preference process on impact. Solid blue lines refer to the D-THANK model and dashed orange lines to the THANK model.

H.2 Fiscal paths following a transfer shock

This appendix reports the fiscal variables underlying the transfer-shock analysis of Section 4.

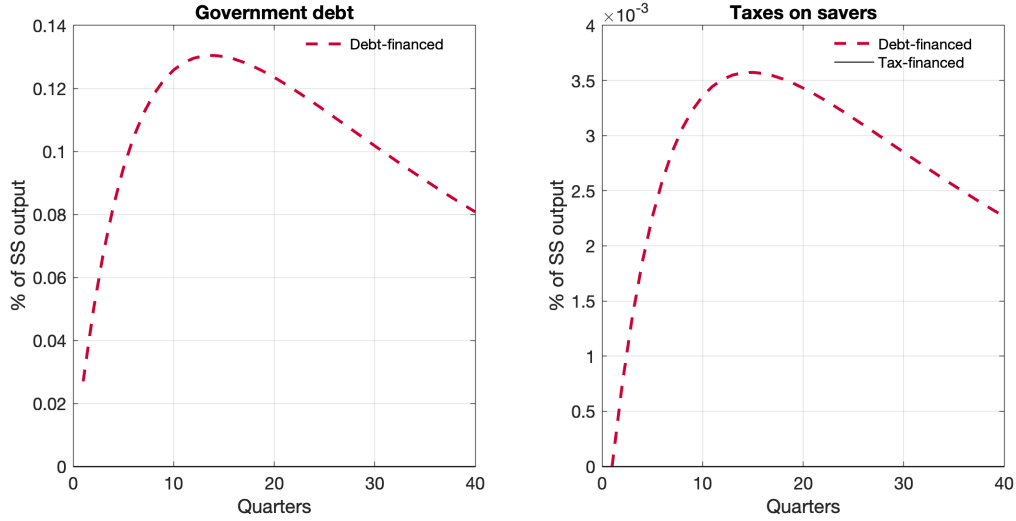


Figure 15: Government debt and saver taxes following a transfer shock

Note: Responses to the transfer shock of Figure 4. The left panel reports the stock of government debt as a share of steady-state output under debt financing; the right panel compares the path of taxes on savers under the two financing regimes. Taxes are reported from the government's perspective, so that positive values denote taxes levied on savers. Under tax financing, taxes offset the transfer contemporaneously; under debt financing, the consolidation rule retires the outstanding stock gradually.

H.3 Targeted versus universal transfers

This appendix compares the targeted transfer policy considered in Section 4, which directs transfers to hand-to-mouth households, with a universal policy that distributes transfers equally across all households. The comparison is conducted under debt financing, holding the aggregate fiscal outlay fixed at 0.1% of steady-state output on impact. With universal transfers, each household receives the same per-capita payment, implying that a fraction λ of the aggregate outlay accrues to hand-to-mouth households.

Figure 16 reports the resulting impulse responses. The portion of the universal transfer received by savers does not affect their consumption decisions, since lump-sum payments do not enter the saver Euler equation and are absorbed intertemporally. As a result, only the fraction of the universal transfer accruing to hand-to-mouth households generates the direct stimulus present under the targeted policy. The responses under the universal design therefore correspond to those under the targeted design scaled by the hand-to-mouth share λ .

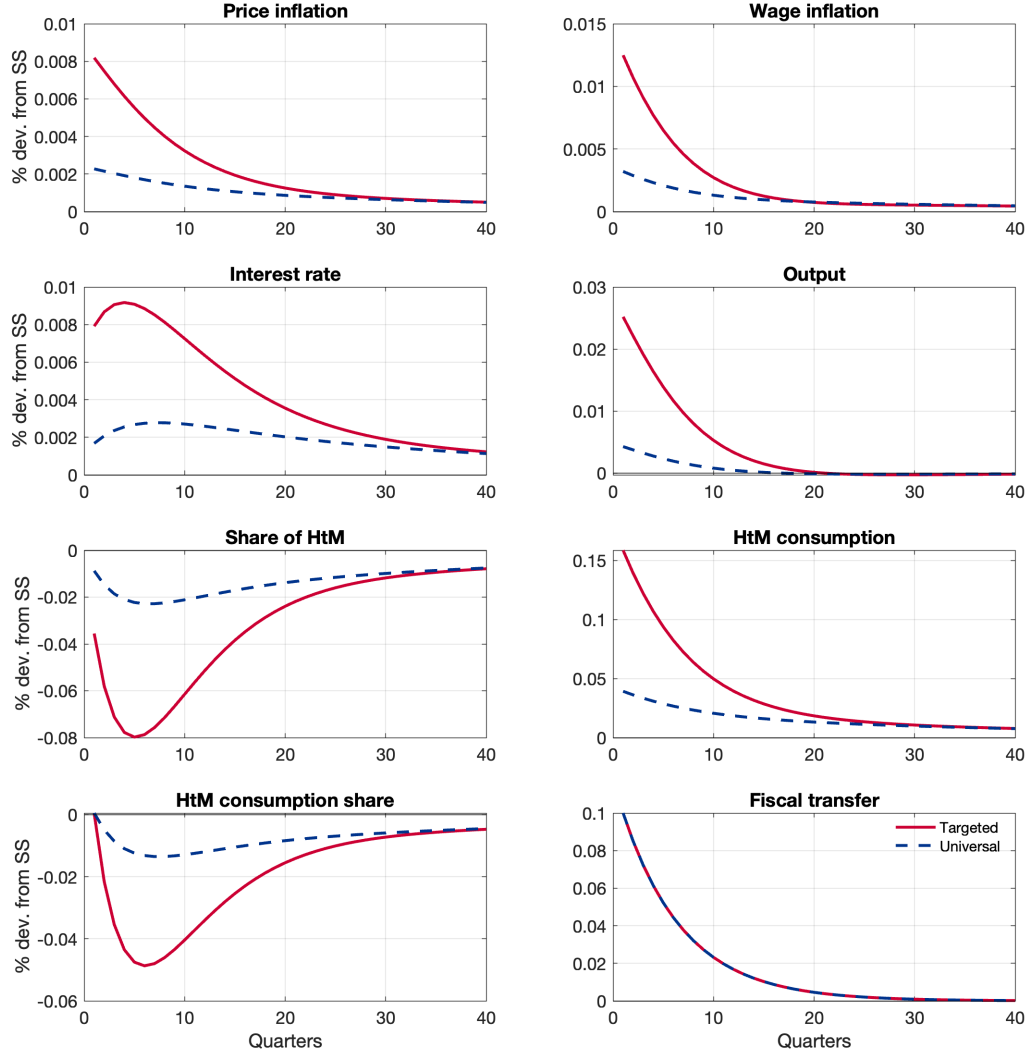


Figure 16: Impulse response functions to a transfer shock under debt financing: targeted versus universal transfers

Note: Impulse responses are constructed at the posterior mode of the structural parameters, under the debt-financing regime of Appendix G. The exogenous transfer shock is normalized to deliver an aggregate outlay of 0.1% of steady-state output on impact, with a persistence parameter of 0.85, under both designs. Solid crimson lines refer to the targeted design and dashed navy lines to the universal design. The consumption share of hand-to-mouth households is constructed from equation (4.5).

The comparison can alternatively be framed by holding fixed the transfer received by each hand-to-mouth household. A universal policy delivering the same payment to constrained households would require a substantially larger aggregate outlay ($1/\lambda$, roughly 3.7 times) and correspondingly greater debt issuance. As can also be inferred from Figure 16, the

larger fiscal expansion would generate a stronger inflationary response while providing the same direct support to hand-to-mouth households. The comparison therefore highlights the greater fiscal efficiency of targeted transfers in directing a given fiscal outlay toward constrained households.

H.4 Optimal transfers: additional impulse responses

This appendix complements the analysis of Section 4. Figure 17 reports the distributional responses to the cost-push shock underlying Figure 6, while Figures 18–21 report the responses to the remaining structural shocks. Each figure compares three policy configurations. Dashed grey lines report the equilibrium without transfers, and dotted orange lines the Ramsey allocation, both as defined in Section 4. Solid blue lines report the estimated rule, under which transfers respond to output and to the consumption share of hand-to-mouth households.

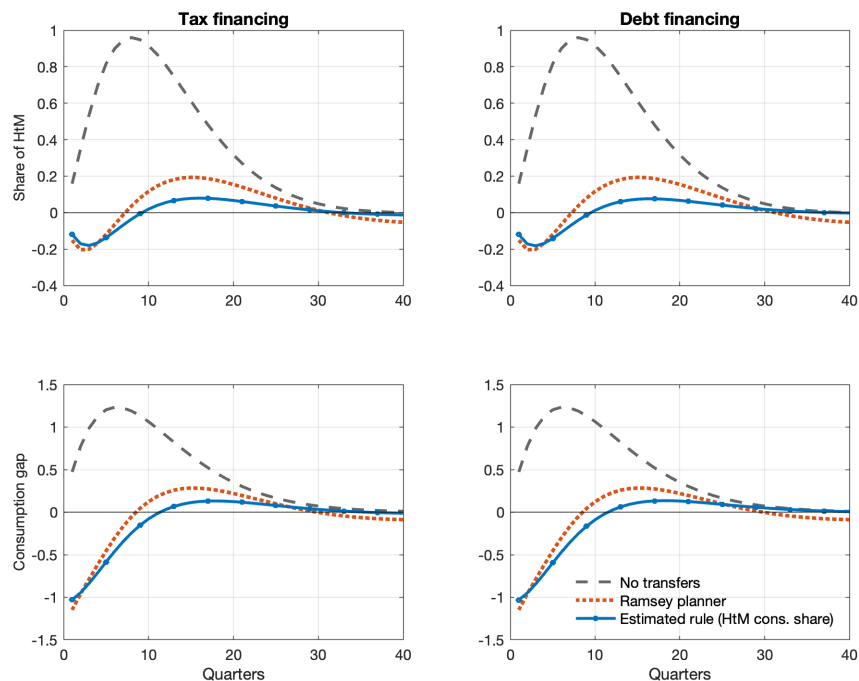


Figure 17: Distributional responses to a cost-push shock under alternative transfer policies
Note: Responses of the share of hand-to-mouth households and the consumption gap to the cost-push shock of Figure 6. Left panels refer to tax financing and right panels to debt financing. Lines follow the conventions described in the text.

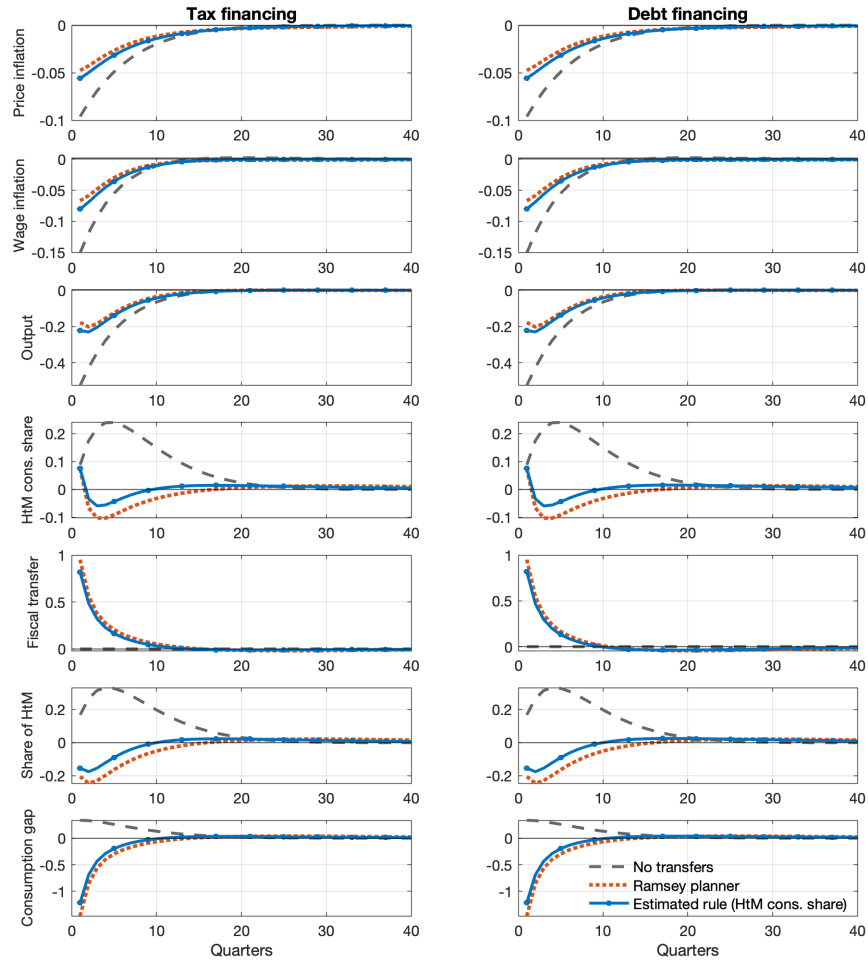


Figure 18: Responses to a monetary policy shock under alternative transfer policies

Note: Impulse responses to a one-standard-deviation monetary policy shock at the posterior mode. Left panels refer to tax financing and right panels to debt financing. Lines follow the conventions described in the text.

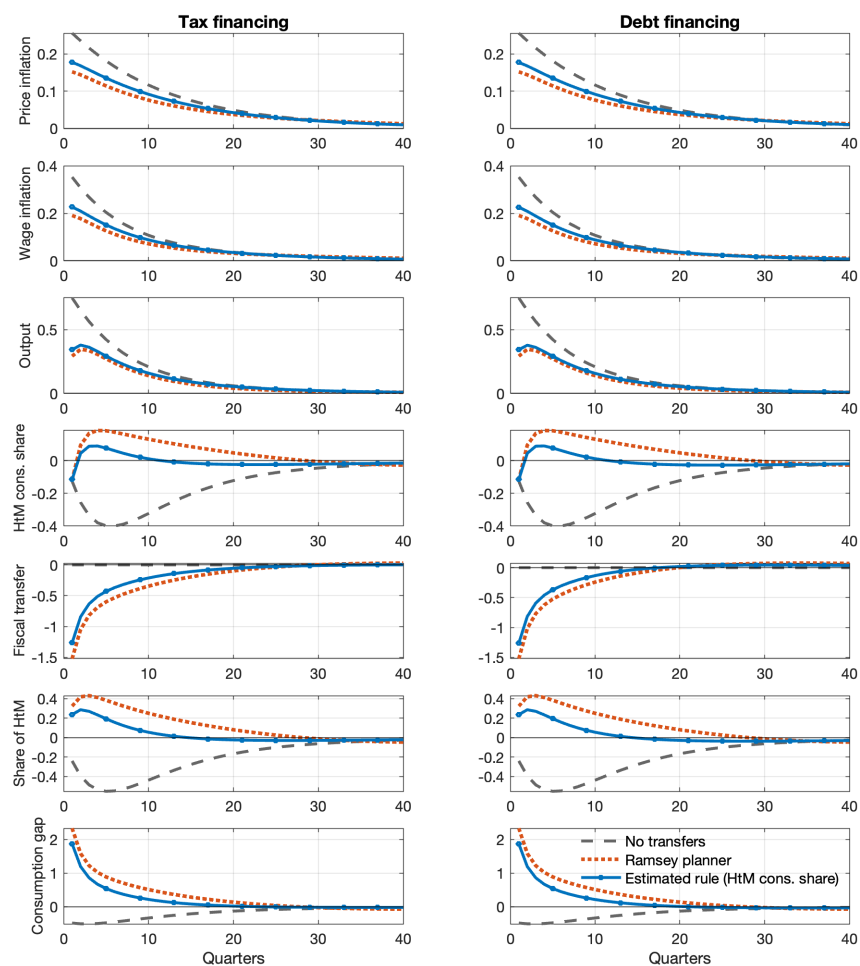


Figure 19: Responses to a preference shock under alternative transfer policies
Note: Impulse responses to a one-standard-deviation preference shock at the posterior mode. Conventions as in Figure 18.

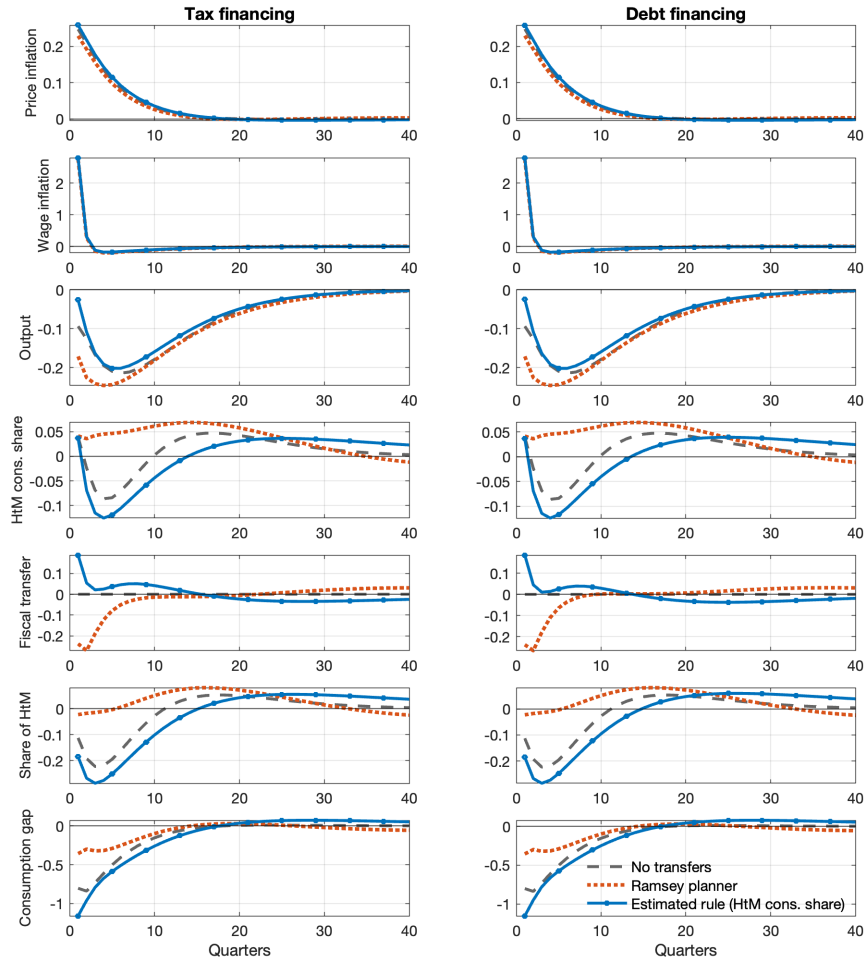


Figure 20: Responses to a wage markup shock under alternative transfer policies
Note: Impulse responses to a one-standard-deviation wage markup shock at the posterior mode. Conventions as in Figure 18.

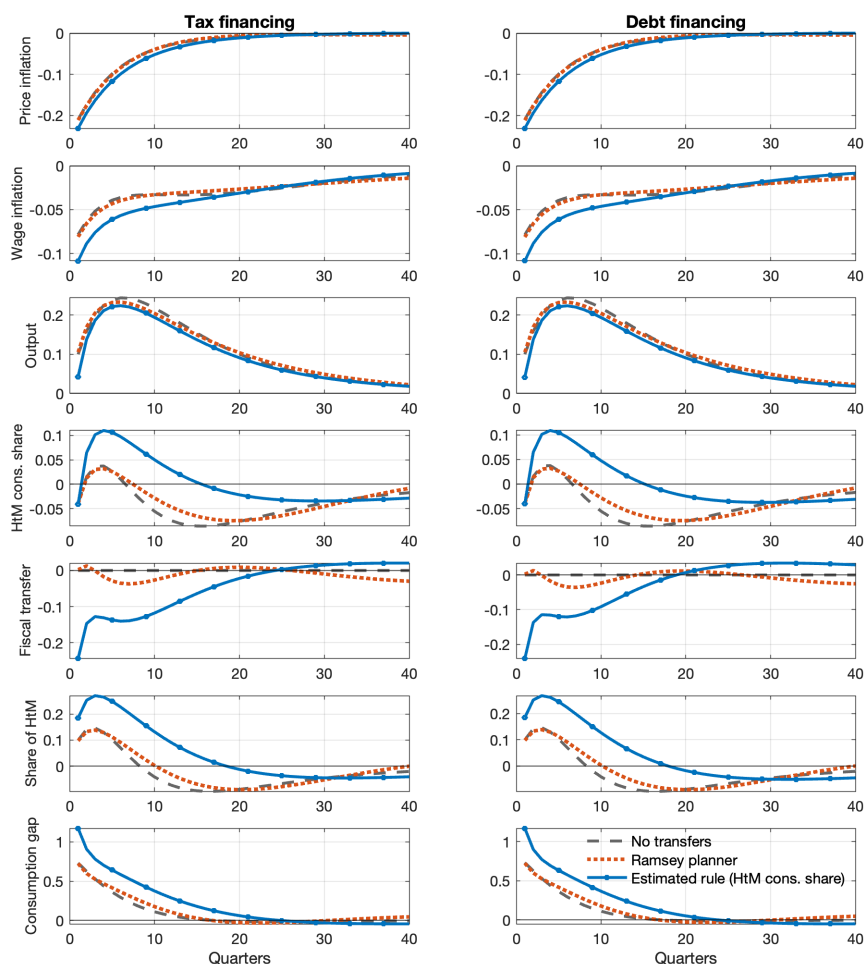


Figure 21: Responses to a technology shock under alternative transfer policies

Note: Impulse responses to a one-standard-deviation technology shock at the posterior mode. Conventions as in Figure 18.

H.5 Fit of the estimated transfer rule

Figure 22 plots the transfers implied by the estimated rule responding to output and to the consumption share of hand-to-mouth households against the planner's transfers along the simulated Ramsey allocation.

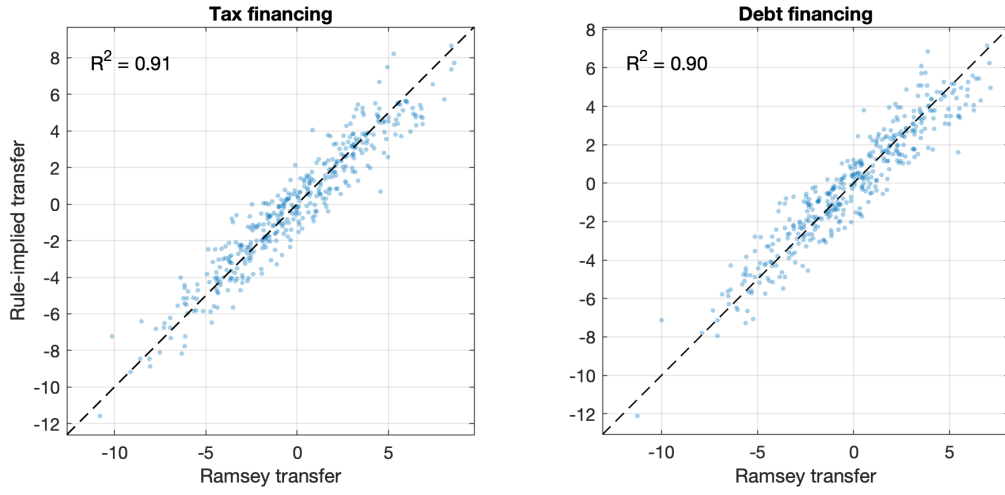


Figure 22: Estimated transfer rule and the Ramsey allocation

Note: Transfers implied by the estimated rule, plotted against the planner's transfers along the simulated Ramsey allocation. The dashed line is the 45-degree line. The estimated coefficients are $\rho_t = 1.00$, $\phi_{y,0} = -2.55$, $\phi_{y,1} = 2.59$, $\phi_{x,0} = 3.33$, $\phi_{x,1} = -3.09$ under tax financing and $\rho_t = 0.99$, $\phi_{y,0} = -2.54$, $\phi_{y,1} = 2.58$, $\phi_{x,0} = 3.39$, $\phi_{x,1} = -3.14$ under debt financing.

I COVID-19 counterfactuals

This appendix provides additional details on the identification of the shocks, the construction of the counterfactuals, and the mechanism decomposition underlying the analysis in Section 5. The data used in the exercise are described in Appendix D.1.

Construction of the counterfactuals. The counterfactual exercise proceeds in two stages. The first stage recovers a combination of preference and productivity disturbances underlying the contraction in economic activity during the pandemic. While the estimated model contains a broader set of structural disturbances and observables, the counterfactual exercise focuses on these two shocks as parsimonious drivers of fluctuations in aggregate demand and supply during the pandemic episode. To identify these “Covid” shocks, I use observables that are less directly affected by the fiscal transfer programs, namely the per-capita growth rate of real personal income excluding transfers and the level of per-capita hours. By contrast, variables such as inflation, nominal wage growth, the policy rate, and transfers are shaped in equilibrium by the fiscal response during the pandemic. Conditioning on these variables would therefore risk incorporating part of the transfer programs being evaluated into the recovered shocks. The sequences $\{\hat{u}_t^a, \hat{u}_t^{pref}\}_{t=2017Q1}^{2022Q1}$ are recovered by applying a calibrated Kalman smoother to the D-THANK model with liquidity (i.e., debt-financed transfers), with all parameters fixed at their posterior modes.

In the second stage, I feed the recovered innovations through alternative configurations of the model, holding the shock sequences fixed and varying only the specification of transfer policy. The no-transfers counterfactual corresponds to the smoothed path obtained in the first stage and is reported by the black dashed lines in Figures 8 and 23. The rule-based counterfactual is obtained by replacing the exogenous transfer process with the estimated transfer rule in equation (4.6), which responds to output and to the consumption share of hand-to-mouth households, the best-performing specification of Section 4.2. Realized outcomes, constructed from the data described in Appendix D.1, are superimposed for com-

parison. All series are reported as deviations from their 2019Q4 levels.

Household composition and self-insurance. How relevant were endogenous household transitions for the distributional and aggregate consequences of the recession? To address this question, I feed the same smoothed shocks through three nested versions of the model. In the first, the switching coefficients are set to zero ($\gamma_s = \gamma_h = \kappa_s = \kappa_h = 0$), holding the hand-to-mouth share fixed at its steady-state value and reducing the model to the standard THANK framework of Bilbiie (2024). In the second, the household distribution is allowed to evolve and enters savers' Euler equation, so that households anticipate the possibility of switching between states, but changes in the distribution do not feed back into aggregate variables. The difference between these two versions isolates the self-insurance mechanism emphasized by Bilbiie et al. (2023). The third corresponds to the full D-THANK model, in which changes in the household distribution also feed back into aggregate variables. The difference between the full model and the second specification therefore isolates the additional effects of endogenous household composition. Figure 23 reports the resulting decomposition.

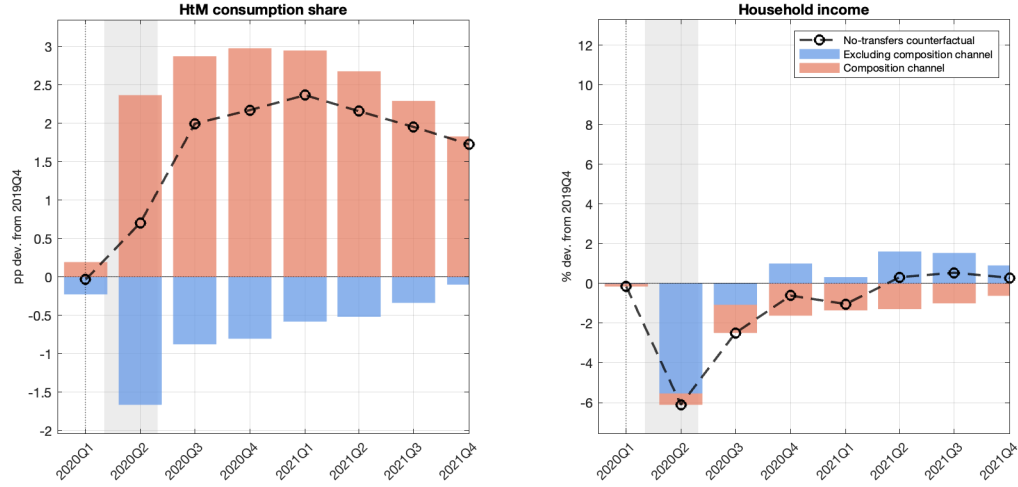


Figure 23: Household composition and self-insurance in the COVID-19 recession

Note: Each panel decomposes the no-transfers counterfactual into the contribution of endogenous changes in household composition and the component excluding composition effects, following the nested model specifications described in the text. The dashed line reports the full no-transfers path, equal to the sum of the two components. The left panel reports the consumption share of hand-to-mouth households, in percentage-point deviations from 2019Q4, and the right panel reports household income, in percent deviations from 2019Q4.

The decomposition highlights the importance of endogenous household transitions for

understanding the distributional consequences of the recession. In the absence of changes in household composition, the relative consumption response of hand-to-mouth households and savers would lead to a decline in the hand-to-mouth share of aggregate consumption. Once endogenous household transitions are taken into account, however, the distributional response changes markedly. The decline in income associated with the COVID-19 shock induces a sharp increase in the fraction of households that become financially constrained. The resulting composition channel more than offsets the relative-consumption channel, raising the hand-to-mouth consumption share to almost 2.5 percentage points above its pre-pandemic level in early 2021 before gradually declining.

A similar, albeit more moderate, pattern emerges for household income. Endogenous household transitions reinforce the initial decline only modestly, but their importance becomes more apparent during the recovery. While the contribution of self-insurance gradually recedes, the composition channel remains negative throughout 2021, continuing to weigh on household income after the initial contraction has subsided. Endogenous household transitions therefore contribute not only to the depth of the recession but also to the persistence of its effects.

These counterfactuals highlight the importance of accounting for endogenous household transitions when assessing the consequences of aggregate shocks. Focusing exclusively on the consumption responses of fixed household types provides an incomplete characterization of distributional dynamics and, as the COVID-19 counterfactual illustrates, can even imply a different direction for the distributional response. Accounting for both margins is consequently important not only for understanding the distributional consequences of recessions, but also for identifying the information most relevant for the design of systematic transfer policies.